Excitation-Current Coordinate Reduction in Electrically Excited Synchronous Machines

Mathematical Foundations, Identification Errors, and Implications for Current Control

25 October 2026

Abstract

An electrically excited synchronous machine generally requires three current coordinates to describe its two stator flux linkages. A compact alternative replaces the physical d-current by a shifted coordinate containing excitation current. This manuscript develops the conditions, identification difficulties and control consequences of that reduction. It begins with a reproducible contour-intersection estimator and derives why a mean contour coordinate is not generally a constant-flux operating point. Cross-saturation, contour curvature, sampling weights and measured excitation scatter are separated mathematically. Exact reduction is characterized by directional derivatives and global fibre consistency; local differential factors, global fitted factors and pointwise flux alignment answer different questions.

Three measured datasets illustrate these distinctions. Conventional factors are approximately 13.49, 13.52 and 14.76 A/A. Their finite-table reconstruction errors, derivative residuals and contour-centroid departures have different units or evaluation populations and must not be interchanged. Pointwise alignment uses the known target flux and is a diagnostic rather than an independent predictor. Two constant axis shifts give an additional reduced-model degree of freedom. Four linear coefficients, however, reduce to two effective excitation factors and an invertible change of the stored coordinates. The observed approximately 0.5% reconstruction improvement from this change is not accompanied by a consistent improvement on one omitted inner excitation group per dataset.

The electrical equations connect flux-map errors to speed-dependent feedforward-voltage errors, differential-inductance errors, PI disturbance rejection and voltage reserve. The controller discussion distinguishes inspected implementations from idealized design assumptions and from an unverified active experimental configuration. Alternative contour statistics, global fitting, two-axis coordinates, full three-current maps and slower auxiliary evaluation are assessed only after the limitations have been established. The conclusions distinguish mathematical exactness, finite numerical approximation and practical usefulness; they do not identify a unique physical source of every observed discrepancy.

Contents

1 Introduction and Motivation **1**

1.1 Why excitation is a third coordinate . . . 1

1.2 The attraction of a smaller table . . . 1

1.3 What a coordinate factor means . . . 2

1.4 Research question and order of argument . . . 2

1.5 Notation and evidence . . . 3

2 Conventional Contour-Based Identification **4**

2.1 What is computed, before interpreting it . . . 4

2.2 Constructing the group meshes . . . 4

2.3 Selecting the flux levels . . . 5

2.4 Finding the edge intersections . . . 5

2.5 Averaging the crossing coordinates . . . 5

2.6 Adjacent differences and the final scalar . . . 6

2.7 Two diagnostic quantities that are not flux-reconstruction errors . . . 6

2.8 What this procedure has and has not identified . . . 7

3 Mathematical Analysis of the Conventional Method **8**

3.1 Contour displacement and differential relations . . . 8

3.1.1 A partial derivative is not a contour trajectory . . . 8

3.1.2 From a differential slope to a finite secant . . . 9

3.2 The displacement of contour averages . . . 9

3.3 Why a mean coordinate need not lie on the contour . . . 10

3.3.1 An exact curved-contour example . . . 10

3.3.2 A local curvature interpretation beyond the example . . . 11

3.4 Dependence on triangulation and sampling . . . 11

3.4.1 Why ratios can be sensitive . . . 12

3.5 Conditions for an exact common scalar reduction . . . 12

3.5.1 The discarded direction . . . 12

3.5.2 The sensitivity-vector geometry . . . 13

3.6 Local and global identification . . . 13
3.6.1 Deriving the local least-squares factor . . . 13
3.6.2 One constant factor fitted to many derivative samples . . . 14
3.6.3 How empirical derivatives enter the argument . . . 15
3.6.4 Why the different factors need not agree . . . 15

4 Sensitivity to Measurement and Modeling Errors 16
4.1 What the stationary voltage calculation assumes . . . 16
4.2 Stator-resistance errors . . . 17
4.2.1 Deriving the component errors and their signs . . . 17
4.2.2 A local d-contour ratio can remain unchanged . . . 17
4.2.3 Why joint scalar reduction can appear inconsistent . . . 17
4.3 Temperature and averaging . . . 18
4.4 Angular errors and measurement delays . . . 18
4.4.1 A constant angle offset rotates both vectors and coordinates . . . 18
4.4.2 A time delay has more than one interpretation . . . 19
4.5 Voltage, current and speed uncertainty . . . 19
4.6 Iron losses and other omitted effects . . . 19
4.7 What can be concluded from these sensitivities . . . 20

5 Experimental Investigation and Reconstruction Errors 21
5.1 Measurement populations and their limitations . . . 21
5.2 From measured fields to a conventional reduced table . . . 22
5.3 Error definitions and coverage . . . 22
5.4 Contour statistics and local derivative evidence . . . 25
5.5 Pointwise scalar alignment as a diagnostic reference . . . 27
5.6 Pointwise two-axis alignment and identifiability . . . 28
5.7 Comparing diagnostic references on identical physical points . . . 29

6 Implications for Current Control 30
6.1 From flux maps to dynamic machine equations . . . 30
6.2 The local current-control plant . . . 31
6.3 What a PI regulator does . . . 32
6.3.1 An ideal magnitude-optimum derivation . . . 33
6.3.2 Verified structures and the limits of their interpretation . . . 34
6.4 Flux feedforward and the meaning of a flux error . . . 34
6.4.1 Connecting coordinate-factor error to voltage sensitivity . . . 36
6.5 How feedback responds to imperfect feedforward . . . 36
6.6 Voltage limitation and anti-windup . . . 37

6.7 Operating regions and practical consequences 38

7 Alternative Identification and Representation Strategies 39

7.1 Robust contour identification 39
7.1.1 Length weighting changes the averaging measure 39
7.1.2 Robust locations and fixed-q comparisons 39
7.2 An optimized global scalar factor 40
7.2.1 Reconstruction fitting and fixed-map alignment are different 40
7.2.2 Keeping the evaluation population fixed 40
7.2.3 What the existing scalar comparisons establish 41
7.3 Two constant global excitation factors 41
7.3.1 The complete local condition 41
7.3.2 Local solutions, global constants and invertibility 42
7.3.3 Consistent global fitting and current results 42
7.3.4 Measured excitation scatter can fold a projected group 43
7.4 A general linear four-parameter transformation 43
7.4.1 Deriving the effective factors 43
7.4.2 Finite grids do not commute with the coordinate change 44
7.4.3 Measured benefit, coverage and a simple prediction check 44
7.5 Full three-dimensional flux maps 45
7.5.1 What retaining excitation removes, and what it does not 45
7.5.2 What the archived prediction results show 46
7.6 Auxiliary computation and reduced online workload 47
7.6.1 Identify the actual dependencies before caching 47
7.6.2 A conditional bound for stale map values 47
7.6.3 Scheduling, consistency and limits 47
7.7 Further possibilities and the boundaries of this evidence 48

8 Discussion and Conclusions 49

8.1 What the conventional procedure assumes 49
8.2 What is mathematically established 49
8.3 What the measured evidence establishes 50
8.4 What follows for current control 50
8.5 Practical choices and remaining questions 51

Chapter 1

Introduction and Motivation

1.1 Why excitation is a third coordinate

In a rotating dq coordinate system, an electrically excited synchronous machine has two stator-current components and an independently supplied rotor excitation current. At a fixed thermal and magnetic state, a quasistatic description therefore has the form

$$\boldsymbol{\psi}(\boldsymbol{i}) = \begin{bmatrix} \psi_d(i_d, i_q, i_e) \\ \psi_q(i_d, i_q, i_e) \end{bmatrix}, \qquad \boldsymbol{i} = \begin{bmatrix} i_d & i_q & i_e \end{bmatrix}^{\mathsf{T}}. \tag{1.1}$$

Each current is a physical measured current in A. Each flux linkage is in Vs. Excitation changes the rotor magnetomotive force; saturation and cross-saturation can change the responses to all three currents. The number of output quantities does not determine the number of independent input coordinates required to describe them.

Equation (1.1) is already a modeling assumption. A current-only map cannot separately represent temperature, hysteretic history, frequency-dependent magnetic effects or time-varying measurement errors. Throughout the mathematical development, these additional variables are either held fixed or absorbed into an empirical approximation. The measured data do not establish an exact memoryless constitutive law.

For a permanent-magnet synchronous machine under fixed magnet conditions, there is no independently commanded excitation current corresponding to i_e. A two-stator-current description can then be natural. An electrically excited machine does not acquire this property merely because its excitation contribution can be represented approximately by a stator-current offset.

1.2 The attraction of a smaller table

Suppose each current axis is represented by n grid nodes and two flux components are stored. A three-dimensional tensor table contains $2n^3$ flux values; a two-dimensional table contains $2n^2$. The count grows by a factor n when the third axis has the same resolution. If its resolution is instead n_e, the factor is n_e. The actual interpolation work depends on the interpolation rule and data layout, not just on these counts.

A common attempt to remove the excitation coordinate introduces

$$s = i_d + k i_e, \qquad \widehat{\boldsymbol{\psi}}(\boldsymbol{i}) = \widehat{\boldsymbol{F}}(s, i_q). \tag{1.2}$$

The coefficient k has units A/A: $k i_e$ must have the units of a stator-current coordinate. The hat denotes an approximation or an identified representation. Equation (1.2) states that a single pair

of flux functions should serve all supported excitation levels after their d-current coordinates have been shifted.

There are two separate questions. First, can the measured fields approximately be represented in these two coordinates? Second, can a finite table, constructed and interpolated in those coordinates, preserve the required accuracy? An exact continuous reduction could still incur table-storage error. Conversely, a scattered interpolant can reproduce distinct training nodes without proving that a smooth, physically consistent reduction exists.

1.3 What a coordinate factor means

The product ki_e is a coordinate offset, not an additional physical stator current flowing through the stator copper. In particular, physical resistive voltage remains $R_s i_d$, not $R_s s$. Excitation losses remain associated with the excitation circuit. The coefficient need not equal a winding distribution factor or a turns ratio.

A turns ratio may motivate an approximate relation between magnetomotive forces in a simplified unsaturated magnetic model. An empirically identified coordinate coefficient additionally reflects which flux component is fitted, the current region, the sampling and interpolation rules, saturation, measurement errors and the optimization objective. Numerical agreement with a turns ratio would require independent physical evidence; it does not follow from the units A/A.

Even exact stator-flux factorization does not make the complete machine equivalent to a permanent-magnet machine. For the usual three-phase amplitude-invariant convention, the electromagnetic torque identity is

$$T_e = \frac{3}{2} p(\psi_d i_q - \psi_q i_d). \tag{1.3}$$

If $\boldsymbol{\psi} = \boldsymbol{F}(s, i_q)$ were exact, substitution of $i_d = s - ki_e$ would give

$$\begin{aligned} T_e &= \frac{3}{2} p\{F_d(s, i_q) i_q - F_q(s, i_q)(s - ki_e)\} \\ &= \frac{3}{2} p\{F_d(s, i_q) i_q - F_q(s, i_q) s + ki_e F_q(s, i_q)\}. \end{aligned} \tag{1.4}$$

The final excitation-dependent term cannot be removed by renaming the table axis. Equation (1.3) also depends on the dq scaling convention; it is not used here as an independent torque calibration.

1.4 Research question and order of argument

The central question is: under what conditions can the two stator flux linkages of an electrically excited synchronous machine be approximated by one common two-coordinate representation, and what limits the identification of those coordinates?

Chapter 2 first defines the conventional contour statistic without changing it. Chapter 3 separates constant-flux differential relations from the movement of numerical contour averages, and derives exactness and identifiability conditions. Chapter 4 then examines possible measurement and modeling errors. These analyses explain why a discrepancy can arise before any alternative fitting strategy is proposed.

Chapter 5 presents measured evidence and explicitly distinguishes reconstruction, omitted-group prediction and pointwise alignment against a fixed map. Chapter 6 develops the consequences for current control. Only Chapter 7 considers alternative identification and representation strategies. Chapter 8 returns to what is established and what remains uncertain.

1.5 Notation and evidence

The first index of a differential inductance identifies the flux output and the second the varied current:

$$L_{xy} = \frac{\partial \psi_x}{\partial i_y}, \qquad x \in \{d, q\}, \quad y \in \{d, q, e\}. \tag{1.5}$$

These derivatives have units H. A differential inductance is a local slope, not generally a ratio ψ_x / i_y. Cross-derivatives are not silently set to zero or forced to be equal. Equality can follow from an appropriate energy description and consistent coordinates under additional assumptions; it is not imposed on separately estimated empirical fields.

The following distinctions govern the interpretation:

- A mathematical identity is conditional on its stated regularity, support and domain assumptions.
- A numerical result belongs to its specified reference map, objective and evaluation population.
- A possible physical explanation is a hypothesis unless independent evidence identifies its contribution.

No complete uncertainty budget or controlled causal separation is available. Differences in flux error are therefore not partitioned into additive amounts of resistance error, temperature error, interpolation error and irreducible physical reduction error.

Chapter 2

Conventional Contour-Based Identification

2.1 What is computed, before interpreting it

The conventional procedure compares average current coordinates associated with the same d-flux level in adjacent nominal excitation groups. It produces one global scalar. It does not track a single physical point at constant q-current, and it does not minimize a joint flux-reconstruction error. This chapter specifies the complete statistic so that its mathematical interpretation can be examined separately.

Let the source records be

$$\mathcal{D} = \{(d_j, q_j, e_j, \psi_{d,j}, \psi_{q,j}, r_j)\}_{j=1}^{N}. \tag{2.1}$$

Here (d_j, q_j, e_j) are measured currents, whereas r_j is a nominal excitation reference used to assign a group. Ordered distinct references define groups $g = 1, \ldots, G$. A nominal reference is not substituted for the measured e_j. In particular, the nominal zero-excitation group need not have exactly zero measured excitation at every record.

2.2 Constructing the group meshes

Within each group, form a Delaunay triangulation in the measured (d, q) plane. Collect the three edges of every triangle, sort each edge's endpoint indices and remove duplicate index pairs. An interior edge shared by two triangles is therefore present once.

For an edge $a-b$, define its squared physical-current length

$$\ell_{ab}^2 = (d_b - d_a)^2 + (q_b - q_a)^2. \tag{2.2}$$

If the unique edge set is $\mathcal{E}_g$, its mean squared length is

$$\overline{\ell^2}_g = \frac{1}{|\mathcal{E}_g|} \sum_{(a,b)\in\mathcal{E}_g} \ell_{ab}^2. \tag{2.3}$$

Only edges with $\ell_{ab}^2 < 9\overline{\ell^2}_g$ are retained. This is an empirical geometry filter in the measured stator-current plane. It is not a rejection based on flux error, and it is not a proof that all remaining edges correspond to physically well-sampled contour pieces.

The factor nine applies to squared length; equivalently, the length is compared with three times the root mean squared edge length. The mean is calculated before this edge rejection. The filter can change which portions of a contour contribute to the subsequent average.

2.3 Selecting the flux levels

Pool the measured d-flux values over all groups, and set

$$\psi_{\min} = \min_j \psi_{d,j}, \qquad \psi_{\max} = \max_j \psi_{d,j}, \qquad \Delta\psi = \psi_{\max} - \psi_{\min}. \tag{2.4}$$

The fitted levels are

$$\lambda_\ell = \psi_{\min} + \left(\frac{1}{4} + \frac{\ell - 1}{99}\frac{1}{2}\right)\Delta\psi, \qquad \ell = 1, \ldots, 100. \tag{2.5}$$

Thus the interval is the middle 25–75% of the *value range*. It is not the interval between empirical flux quantiles. Uneven sample density does not change this distinction.

The same levels are used for every group. Some levels can lack intersections in some groups; the procedure does not fabricate those contours. Additional 201 levels spanning the full pooled range can be displayed for diagnosis, but they do not enter the fitted scalar. Values outside the fitted interval in such a display therefore illustrate behavior without changing the estimator.

2.4 Finding the edge intersections

Along a retained edge, assume linear interpolation of the recorded d-flux:

$$\widetilde{\psi}_d(t) = (1 - t)\psi_{d,a} + t\psi_{d,b}, \qquad 0 \le t \le 1. \tag{2.6}$$

Setting this expression equal to λ_ℓ and subtracting $\psi_{d,a}$ gives

$$t(\psi_{d,b} - \psi_{d,a}) = \lambda_\ell - \psi_{d,a}, \tag{2.7}$$

$$t_{ab,\ell} = \frac{\lambda_\ell - \psi_{d,a}}{\psi_{d,b} - \psi_{d,a}}. \tag{2.8}$$

A crossing is retained only if this computed fraction lies in $[0, 1]$. Equal endpoint fluxes produce a zero denominator; nonfinite fractions do not satisfy this condition. In particular, a complete edge coinciding with the level is not integrated as a contour segment by this rule.

Interpolate all three measured currents with the same fraction:

$$\boldsymbol{c}_{ab,\ell} = \begin{bmatrix} d_a \\ q_a \\ e_a \end{bmatrix} + t_{ab,\ell}\left(\begin{bmatrix} d_b \\ q_b \\ e_b \end{bmatrix} - \begin{bmatrix} d_a \\ q_a \\ e_a \end{bmatrix}\right). \tag{2.9}$$

Although the mesh has only two coordinate axes, excitation is an interpolated measured scalar attached to its vertices. A nominal group therefore represents an empirical excitation graph over (d, q), not necessarily a plane of exactly constant measured e.

2.5 Averaging the crossing coordinates

If group g and level ℓ have $M_{g\ell}$ retained crossings, calculate

$$\overline{\boldsymbol{c}}_{g\ell} = \frac{1}{M_{g\ell}} \sum_{m=1}^{M_{g\ell}} \boldsymbol{c}_{g\ell,m} = \begin{bmatrix} \bar{d}_{g\ell} \\ \bar{q}_{g\ell} \\ \bar{e}_{g\ell} \end{bmatrix}. \tag{2.10}$$

An empty crossing set yields an unavailable mean. Every crossing has equal weight. The construction does not weight a point by the length of the contour segment it represents or by measurement uncertainty.

Removing duplicate edges does not remove every geometrically repeated crossing. If a contour passes exactly through a vertex, several distinct incident edges can contribute that vertex. This multiplicity is part of the arithmetic statistic. Disconnected contour pieces are pooled unless the geometry or edge filter excludes them. The term *centroid* below is shorthand for the coordinate average (2.10); it does not imply an area centroid or an arclength centroid.

2.6 Adjacent differences and the final scalar

For adjacent ordered groups, form

$$\Delta\bar{d}_{g\ell} = \bar{d}_{g+1,\ell} - \bar{d}_{g\ell}, \tag{2.11}$$

$$\Delta\bar{q}_{g\ell} = \bar{q}_{g+1,\ell} - \bar{q}_{g\ell}, \tag{2.12}$$

$$\Delta\bar{e}_{g\ell} = \bar{e}_{g+1,\ell} - \bar{e}_{g\ell}. \tag{2.13}$$

The conventional individual ratio and a separate signed diagnostic are

$$\kappa_{g\ell} = \frac{|\Delta\bar{d}_{g\ell}|}{\Delta\bar{e}_{g\ell}}, \qquad \kappa_{g\ell}^{\text{signed}} = -\frac{\Delta\bar{d}_{g\ell}}{\Delta\bar{e}_{g\ell}}. \tag{2.14}$$

Only the numerator of κ is absolute. The denominator remains signed. Sorting the complete nominal groups is therefore essential, but it does not mathematically force every measured centroid denominator to be positive. If a measured denominator is negative, a finite negative ratio remains negative. No positive-factor constraint is inserted into the conventional mean.

Let $\mathcal{I}$ contain the finite ratios from the 100 fitted levels whose magnitudes are not below the numerical machine-precision threshold used by the procedure. Then

$$\boxed{k_{\text{conv}} = \frac{1}{|\mathcal{I}|} \sum_{(g,\ell)\in\mathcal{I}} \frac{|\bar{d}_{g+1,\ell} - \bar{d}_{g\ell}|}{\bar{e}_{g+1,\ell} - \bar{e}_{g\ell}}.} \tag{2.15}$$

The numerical zero screen is not a physical uncertainty criterion. A very small but nonzero denominator can still create a very large finite contribution.

Equation (2.15) is a mean of quotients, not a quotient of pooled displacements:

$$\frac{1}{m}\sum_{n=1}^{m}\frac{A_n}{B_n} \neq \frac{\sum_n A_n}{\sum_n B_n} \tag{2.16}$$

in general. A group pair and level receive one equal contribution regardless of the number of crossings or the excitation separation. These implicit weights matter when interpreting the final factor.

2.7 Two diagnostic quantities that are not flux-reconstruction errors

The q-current drift of adjacent contour averages is

$$D_{g\ell} = \frac{\Delta\bar{q}_{g\ell}}{\Delta\bar{e}_{g\ell}}. \tag{2.17}$$

It measures movement of the average q-coordinate per excitation-current change. It does not measure the response of a point constrained to a specified physical q-current.

Construct a separate linear, nonextrapolating interpolation $\widetilde{\psi}_{d,g}(d, q)$ from the group's original source values. The centroid departure is

$$\epsilon_{g\ell}^{\text{cent}} = \widetilde{\psi}_{d,g}(\bar{d}_{g\ell}, \bar{q}_{g\ell}) - \lambda_{\ell}. \tag{2.18}$$

It asks whether the average coordinate lies on the chosen contour of that group interpolant. The centroid interpolation uses the original group triangulation; its evaluation is not the result of querying a merged reduced lookup table. Its support rule also need not coincide with the filtered crossing-edge rule.

A large ϵ^{cent} proves a departure from the chosen interpolated contour at that average coordinate. It does not directly equal an error in k_{conv} or a reconstructed-flux error. Chapter 3 shows precisely why differences of centroid biases, rather than their absolute sizes alone, affect a displacement estimate.

2.8 What this procedure has and has not identified

The procedure is now reproducible: measured currents determine the geometry and interpolated coordinates; nominal references define the group order; 100 pooled levels determine the fitted range; equal crossing weights and equal quotient weights determine the statistic. None of these steps uses ψ_q to choose k_{conv}.

An interpretation as a local differential compensation factor requires additional assumptions about the path, q-current, centroid location and local sensitivities. An interpretation as an optimal two-output reduction additionally requires a specified reconstruction objective. Neither interpretation follows simply from the name of the coefficient or from smooth-looking shifted surfaces.

Chapter 3

Mathematical Analysis of the Conventional Method

3.1 Contour displacement and differential relations

3.1.1 A partial derivative is not a contour trajectory

Assume first that ψ_d is continuously differentiable on an open current domain. Changing only excitation while holding both stator currents fixed gives

$$\left.\frac{\partial \psi_d}{\partial i_e}\right|_{i_d, i_q} = L_{de}. \tag{3.1}$$

This experiment generally changes d-flux. A constant-flux displacement asks a different question: how must other currents move to compensate that change?

Parameterize a differentiable path by τ, with currents $i_d(\tau), i_q(\tau), i_e(\tau)$. Applying the chain rule to the composite function gives

$$\frac{\mathrm{d}\psi_d}{\mathrm{d}\tau} = L_{dd}\frac{\mathrm{d}i_d}{\mathrm{d}\tau} + L_{dq}\frac{\mathrm{d}i_q}{\mathrm{d}\tau} + L_{de}\frac{\mathrm{d}i_e}{\mathrm{d}\tau}. \tag{3.2}$$

The total derivative includes every input that changes along the path. A partial derivative is one special choice of path, with the other input derivatives zero.

Suppose d-flux is constant, $\psi_d = \lambda$, and $\mathrm{d}i_e/\mathrm{d}\tau \neq 0$ so that excitation itself can be used as a local path parameter. Divide (3.2) by $\mathrm{d}i_e/\mathrm{d}\tau$:

$$0 = L_{dd}\frac{\mathrm{d}i_d}{\mathrm{d}i_e} + L_{dq}\frac{\mathrm{d}i_q}{\mathrm{d}i_e} + L_{de}, \tag{3.3}$$

$$L_{dd}\frac{\mathrm{d}i_d}{\mathrm{d}i_e} = -L_{de} - L_{dq}\frac{\mathrm{d}i_q}{\mathrm{d}i_e}. \tag{3.4}$$

If $L_{dd} \neq 0$, division and reversal of the sign yield

$$\boxed{-\frac{\mathrm{d}i_d}{\mathrm{d}i_e} = \frac{L_{de}}{L_{dd}} + \frac{L_{dq}}{L_{dd}}\frac{\mathrm{d}i_q}{\mathrm{d}i_e}.} \tag{3.5}$$

The first term is excitation compensation by d-current at fixed q-current. The second is additional compensation for whatever change in d-flux is caused by the moving q-current. Cross-saturation enters through L_{dq}; a moving q-coordinate alone is not enough if this derivative vanishes.

At fixed i_q, $\mathrm{d}i_q/\mathrm{d}i_e = 0$, and

$$-\left.\frac{\mathrm{d}i_d}{\mathrm{d}i_e}\right|_{\psi_d, i_q} = \frac{L_{de}}{L_{dd}}. \tag{3.6}$$

The implicit-function theorem supplies a local solution for i_d under the nonzero-L_{dd} assumption. It does not guarantee that the same contour branch extends through the entire measured region. Near a vanishing denominator, a d-current compensation slope can become ill-conditioned or cease to be a suitable parameterization.

3.1.2 From a differential slope to a finite secant

For a single admissible constant-flux path connecting excitation values e_1, e_2, integrate the derivative:

$$i_d(e_2) - i_d(e_1) = \int_{e_1}^{e_2} \frac{\mathrm{d}i_d}{\mathrm{d}e}\,\mathrm{d}e, \tag{3.7}$$

$$-\frac{\Delta i_d}{\Delta i_e} = \frac{1}{e_2 - e_1}\int_{e_1}^{e_2}\left(\frac{L_{de}}{L_{dd}} + \frac{L_{dq}}{L_{dd}}\frac{\mathrm{d}i_q}{\mathrm{d}e}\right)\mathrm{d}e. \tag{3.8}$$

A finite quotient is therefore a path average, even when both endpoints really lie on the same contour. It equals one fixed local ratio only under further conditions, for example constant sensitivities and fixed q-current. Applying (3.8) to arbitrary numerical centroids would assume precisely what must first be established: that their flux values are equal and that an admissible connecting contour path exists.

3.2 The displacement of contour averages

The contour statistic changes the whole crossing population when the excitation group changes. Some crossing coordinates move physically; others appear or disappear because a contour is clipped by the measurement boundary or intersects a different mesh. Thus $\Delta\bar{i}_q/\Delta\bar{i}_e$ mixes a geometric displacement with changes of the averaging population.

For a hypothetical differentiable family of centroid coordinates, parameterized by its mean excitation e, define the full-field value

$$\chi(e) = \psi_d(\bar{d}(e), \bar{q}(e), e). \tag{3.9}$$

The same chain rule now gives

$$\chi'(e) = L_{dd}\bar{d}'(e) + L_{dq}\bar{q}'(e) + L_{de}. \tag{3.10}$$

Rearranging shows the missing term in a constant-flux interpretation:

$$-\bar{d}'(e) = \frac{L_{de}}{L_{dd}} + \frac{L_{dq}}{L_{dd}}\bar{q}'(e) - \frac{\chi'(e)}{L_{dd}}. \tag{3.11}$$

The drift term is only one contribution. If the average leaves the contour and its flux value changes, χ' contributes as well. The measured diagnostic (2.18) evaluates a group interpolant, not an independently measured full three-current field at the centroid. It cannot automatically be substituted for $\chi - \lambda$ in (3.11).

An exact finite-endpoint identity is also instructive. Let $\boldsymbol{c}_0, \boldsymbol{c}_1$ be two centroid coordinates and assume the straight segment $\boldsymbol{c}(t) = \boldsymbol{c}_0 + t\Delta\boldsymbol{c}$ lies inside a differentiable full-field domain. Then

$$\Delta\psi_d = \int_0^1 \left(L_{dd}(\boldsymbol{c}(t))\Delta\bar{d} + L_{dq}(\boldsymbol{c}(t))\Delta\bar{q} + L_{de}(\boldsymbol{c}(t))\Delta\bar{e}\right)\mathrm{d}t. \tag{3.12}$$

Writing $\langle L_{xy}\rangle = \int_0^1 L_{xy}(\boldsymbol{c}(t))\mathrm{d}t$, and assuming $\Delta\bar{e} \neq 0$ and $\langle L_{dd}\rangle \neq 0$, gives

$$-\frac{\Delta\bar{d}}{\Delta\bar{e}} = \frac{\langle L_{de}\rangle}{\langle L_{dd}\rangle} + \frac{\langle L_{dq}\rangle}{\langle L_{dd}\rangle}\frac{\Delta\bar{q}}{\Delta\bar{e}} - \frac{\Delta\psi_d}{\langle L_{dd}\rangle\Delta\bar{e}}. \tag{3.13}$$

This identity does not assert that the segment is a contour. The last term accounts for its possibly unequal endpoint fluxes. A ratio of path-averaged derivatives in (3.13) and an average of derivative ratios along a contour in (3.8) are different mathematical objects.

3.3 Why a mean coordinate need not lie on the contour

3.3.1 An exact curved-contour example

Consider the analytic field

$$\psi_d(d, q, e) = L_d(d + k_0 e) + cq^2, \qquad L_d > 0. \tag{3.14}$$

Here L_d is a constant in H, k_0 is in A/A, and c is in Vs/A^2. The field has an exact scalar reduction with coefficient k_0. Its d-flux contours at level λ satisfy

$$L_d(d + k_0 e) = \lambda - cq^2, \tag{3.15}$$

$$d = \frac{\lambda}{L_d} - k_0 e - \frac{c}{L_d}q^2. \tag{3.16}$$

The contour is curved in the stator-current plane when $c \neq 0$. Average (3.16) over any finite set of exact contour points, including measured variation in e:

$$\bar{d} = \frac{\lambda}{L_d} - k_0\bar{e} - \frac{c}{L_d}\overline{q^2}, \qquad \overline{q^2} = \frac{1}{M}\sum_m q_m^2. \tag{3.17}$$

Now evaluate the original field at the average coordinate:

$$\psi_d(\bar{d}, \bar{q}, \bar{e}) = L_d\left(\frac{\lambda}{L_d} - \frac{c}{L_d}\overline{q^2}\right) + c\bar{q}^2 \tag{3.18}$$

$$= \lambda - c(\overline{q^2} - \bar{q}^2). \tag{3.19}$$

With population variance $\mathrm{var}(q) = \overline{q^2} - \bar{q}^2$, the centroid departure is exactly

$$\boxed{\psi_d(\bar{d}, \bar{q}, \bar{e}) - \lambda = -c\ \mathrm{var}(q).} \tag{3.20}$$

For $c > 0$, the centroid lies below the chosen flux level. Even a symmetric crossing population with $\bar{q} = 0$ has a nonzero departure whenever its q-coordinates have nonzero spread. Vanishing mean q-current therefore does not eliminate curvature bias.

For two groups evaluated at the same λ, subtract (3.17):

$$\Delta\bar{d} = -k_0\Delta\bar{e} - \frac{c}{L_d}\Delta\overline{q^2}. \tag{3.21}$$

Hence the signed displacement ratio is

$$-\frac{\Delta\bar{d}}{\Delta\bar{e}} = k_0 + \frac{c}{L_d}\frac{\Delta\overline{q^2}}{\Delta\bar{e}}. \tag{3.22}$$

The second moment decomposes as

$$\Delta\overline{q^2} = \Delta\,\mathrm{var}(q) + \Delta(\bar{q}^2). \tag{3.23}$$

Both a change of mean position and a change of spread can alter the estimated displacement. If both groups have zero mean q-current, only the variance difference remains. If their second moments are equal, the signed factor is exactly k_0 despite a potentially large, identical centroid departure in both groups. A centroid error is therefore not automatically an equally large factor error. The absolute numerator in the conventional estimator requires a further sign check before (3.22) can be identified with its individual ratio.

3.3.2 A local curvature interpretation beyond the example

Let $\boldsymbol{\mu} = M^{-1}\sum_m \boldsymbol{c}_m$ and $\boldsymbol{\delta}_m = \boldsymbol{c}_m - \boldsymbol{\mu}$. For a sufficiently smooth scalar field,

$$\psi_d(\boldsymbol{c}_m) = \psi_d(\boldsymbol{\mu}) + \nabla\psi_d(\boldsymbol{\mu})^{\mathsf{T}}\boldsymbol{\delta}_m + \frac{1}{2}\boldsymbol{\delta}_m^{\mathsf{T}} H_d(\boldsymbol{\mu})\boldsymbol{\delta}_m + R_m, \tag{3.24}$$

where H_d is the Hessian. Average this expression. The linear term vanishes because $\sum_m \boldsymbol{\delta}_m = 0$. If all points have exact flux λ and $\Sigma = M^{-1}\sum_m \boldsymbol{\delta}_m\boldsymbol{\delta}_m^{\mathsf{T}}$, then

$$\psi_d(\boldsymbol{\mu}) - \lambda = -\frac{1}{2}\operatorname{tr}(H_d(\boldsymbol{\mu})\Sigma) - \overline{R}. \tag{3.25}$$

For bounded third derivatives over a sufficiently small neighborhood, R_m is of third order in $\|\boldsymbol{\delta}_m\|$. Curvature and the spatial distribution of points act together. A large contour, disconnected pieces or strong variation of curvature can invalidate a local second-order approximation, even though the exact parabolic example remains valid for any spread.

3.4 Dependence on triangulation and sampling

The unweighted average is an average with respect to an empirical crossing measure, not an invariant property of the contour geometry. To see this without any machine model, suppose a crossing set has mean $\boldsymbol{\mu}$ and size M. Add one duplicate of an existing crossing $\boldsymbol{c}_r$. Its new mean is

$$\boldsymbol{\mu}' = \frac{M\boldsymbol{\mu} + \boldsymbol{c}_r}{M+1}, \qquad \boldsymbol{\mu}' - \boldsymbol{\mu} = \frac{\boldsymbol{c}_r - \boldsymbol{\mu}}{M+1}. \tag{3.26}$$

The geometric contour has not changed, but its arithmetic mean changes unless $\boldsymbol{c}_r = \boldsymbol{\mu}$. Refining the mesh in one region can have this same weighting effect by creating more intersections there.

If crossings approach a smooth contour parameterized by arclength ℓ with nonuniform asymptotic density $\rho(\ell)$, their arithmetic mean approaches

$$\frac{\int \boldsymbol{c}(\ell)\rho(\ell)\,\mathrm{d}\ell}{\int \rho(\ell)\,\mathrm{d}\ell}. \tag{3.27}$$

An arclength centroid would instead set $\rho = 1$. Equal geometric contours can therefore yield different arithmetic limits under different sampling densities. Disconnected pieces, vertex multiplicity and truncated contours create additional effective weights.

Delaunay connectivity depends on the metric of the triangulation coordinates. Uniform rotation and scaling preserve its Euclidean geometric criterion in the usual nondegenerate case; independent axis rescaling need not do so. Near co-circular configurations, small coordinate changes can switch a diagonal. The intersection population and filtered edges can then change even when the underlying continuous field does not. The conventional triangulation uses physical stator-current axes; the common reduced table uses range-normalized transformed coordinates. These are different numerical operations.

The long-edge rule limits some sparse connections but can preferentially remove one side of a curved or asymmetric contour. No symmetry completion is applied. Consequently, observed q-drift may combine genuine contour movement and discretization or coverage effects. The available statistics do not uniquely apportion these contributions.

3.4.1 Why ratios can be sensitive

For a differentiable ratio $r = A/B$ near $B \neq 0$,

$$\delta r \simeq \frac{\delta A}{B} - \frac{A}{B^2}\delta B, \tag{3.28}$$

so the first-order variance estimate is

$$\operatorname{var}(r) \simeq \frac{\sigma_A^2}{B^2} + \frac{A^2\sigma_B^2}{B^4} - \frac{2A}{B^3}\operatorname{cov}(A,B). \tag{3.29}$$

Small excitation separations amplify both numerator and denominator uncertainty. Averaging many ratios does not remove a systematic displacement bias shared by them.

Even zero-mean errors in A and B need not yield an unbiased ratio. A second-order expansion around nonzero mean μ_B gives

$$\mathbb{E}[A/B] \simeq \frac{\mu_A}{\mu_B} - \frac{\operatorname{cov}(A,B)}{\mu_B^2} + \frac{\mu_A \operatorname{var}(B)}{\mu_B^3}. \tag{3.30}$$

These local expansions assume fluctuations small compared with $|\mu_B|$. They do not justify a Gaussian uncertainty model near a denominator crossing zero. The conventional absolute numerator is also nondifferentiable at zero displacement, so its uncertainty needs separate treatment there.

3.5 Conditions for an exact common scalar reduction

3.5.1 The discarded direction

Let $\Omega \subset \mathbb{R}^3$ be open and $\boldsymbol{\psi} \in C^1(\Omega)$. For a prescribed constant k, define

$$\pi_k(i_d, i_q, i_e) = (i_d + k i_e, i_q). \tag{3.31}$$

An exact reduction requires one function $\boldsymbol{F}$ such that

$$\boldsymbol{\psi} = \boldsymbol{F} \circ \pi_k. \tag{3.32}$$

All physical points with the same (s, i_q) lie on a *fibre*. Moving on that fibre changes excitation by $\mathrm{d}i_e$ and changes d-current by $-k\,\mathrm{d}i_e$, with q-current fixed.

Differentiate the two output components of (3.32):

$$L_{dd} = F_{d,s}, \qquad L_{de} = kF_{d,s}, \tag{3.33}$$

$$L_{qd} = F_{q,s}, \qquad L_{qe} = kF_{q,s}. \tag{3.34}$$

Both outputs must therefore satisfy

$$\boxed{L_{de} = kL_{dd}, \qquad L_{qe} = kL_{qd}.} \tag{3.35}$$

Identifying a slope from d-flux alone verifies at most the first relation. The second need not follow from it.

Proposition 3.1. *For a fixed constant k, conditions (3.35) are necessary and locally sufficient for (3.32). They are globally sufficient if every nonempty fibre in Ω is connected. If fibre components are disconnected, equality of flux between those components is an additional requirement.*

Proof. Use the invertible change of coordinates $(i_d, i_q, i_e) = (s - ke, q, e)$ and define $\boldsymbol{h}(s, q, e) = \boldsymbol{\psi}(s - ke, q, e)$. Its derivative at fixed (s, q) is

$$\frac{\partial \boldsymbol{h}}{\partial e} = \begin{bmatrix} L_{de} - kL_{dd} \\ L_{qe} - kL_{qd} \end{bmatrix}. \tag{3.36}$$

Necessity follows because a function of (s, q) alone cannot vary with e. Conversely, if this derivative is zero on an open neighborhood, $\boldsymbol{h}$ is locally independent of e, giving a local $\boldsymbol{F}$. Along a connected fibre the fundamental theorem of calculus makes all these values equal. Defining $\boldsymbol{F}$ by that common value is well posed; local coordinate neighborhoods give its C^1 regularity on the open projected domain. Zero derivative on one disconnected interval says nothing about the constant value on another interval, which proves the final qualification. □

Connectedness of the whole current domain does not by itself ensure connected fibres. The proposition also requires the *same* k throughout the domain. Finding some suitable local factor at each point is a weaker statement.

3.5.2 The sensitivity-vector geometry

Define the two output sensitivities to d-current and excitation:

$$\boldsymbol{a} = \begin{bmatrix} L_{dd} \\ L_{qd} \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} L_{de} \\ L_{qe} \end{bmatrix}. \tag{3.37}$$

Exact local scalar compensation requires $\boldsymbol{b} = k\boldsymbol{a}$. If $\boldsymbol{a} \neq 0$, such a scalar exists exactly when

$$\Delta = L_{dd}L_{qe} - L_{qd}L_{de} = 0. \tag{3.38}$$

The determinant is the signed area spanned by these vectors in two-output sensitivity space; its units are H^2. If both vectors are nonzero, normalize it:

$$\eta = \frac{|\Delta|}{\|\boldsymbol{a}\| \, \|\boldsymbol{b}\|} = |\sin\theta|. \tag{3.39}$$

This measures directional mismatch, not relative flux error. If $\boldsymbol{a} = 0$ and $\boldsymbol{b} \neq 0$, no d-shift compensates excitation. If both vanish, every scalar satisfies the local equation and the factor is unidentifiable. A vanishing determinant alone would fail to distinguish these cases.

3.6 Local and global identification

3.6.1 Deriving the local least-squares factor

When $\boldsymbol{b}$ is not parallel to $\boldsymbol{a}$, minimize its unexplained sensitivity:

$$E(k) = \|\boldsymbol{b} - k\boldsymbol{a}\|^2 \tag{3.40}$$

$$= \boldsymbol{b}^{\mathsf{T}}\boldsymbol{b} - 2k\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b} + k^2\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}. \tag{3.41}$$

Differentiating and setting the result to zero gives

$$E'(k) = -2\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b} + 2k\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a} = 0, \tag{3.42}$$

$$k_{\mathrm{loc}} = \frac{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathsf{T}}\boldsymbol{a}}, \qquad \boldsymbol{a} \neq 0. \tag{3.43}$$

The positive second derivative $2\|\boldsymbol{a}\|^2$ establishes the unique minimum under this nonzero-sensitivity assumption. The residual $\boldsymbol{r} = \boldsymbol{b} - k_{\mathrm{loc}}\boldsymbol{a}$ satisfies $\boldsymbol{a}^{\mathsf{T}}\boldsymbol{r} = 0$: the remaining excitation response is perpendicular to the available compensation direction.

For two components, direct expansion further gives

$$\min_k \|\boldsymbol{b} - k\boldsymbol{a}\|^2 = \frac{\Delta^2}{\|\boldsymbol{a}\|^2}. \tag{3.44}$$

This connects determinant and LS residual without identifying either as a finite flux-reconstruction error.

The separate ratios L_{de}/L_{dd} and L_{qe}/L_{qd} each fit one component exactly when their denominator is nonzero. They can disagree. The joint formula (3.43) avoids division by an individually small cross-inductance, but still becomes poorly conditioned when the whole vector $\boldsymbol{a}$ is small.

3.6.2 One constant factor fitted to many derivative samples

For a fixed derivative population $\mathcal{V}$, minimize

$$E_{\mathrm{glob}}(k) = \sum_{j\in\mathcal{V}} \|\boldsymbol{b}_j - k\boldsymbol{a}_j\|^2 . \tag{3.45}$$

The same differentiation yields

$$k_{\mathrm{der}} = \frac{\sum_j \boldsymbol{a}_j^{\mathsf{T}}\boldsymbol{b}_j}{\sum_j \|\boldsymbol{a}_j\|^2} = \frac{\sum_j \|\boldsymbol{a}_j\|^2 k_{\mathrm{loc},j}}{\sum_j \|\boldsymbol{a}_j\|^2}. \tag{3.46}$$

Thus the global derivative factor is a sensitivity-weighted mean of local factors, not their arithmetic mean or median. Changing the retained population or current weighting changes the fitted global value.

The reported relative derivative residual has the form

$$\rho(k) = \frac{\left(\sum_j \|\boldsymbol{b}_j - k\boldsymbol{a}_j\|^2\right)^{1/2}}{\left(\sum_j \|\boldsymbol{b}_j\|^2\right)^{1/2}}. \tag{3.47}$$

The raw residual is in H; ρ is dimensionless. Neither is an error in Vs. Along an actual discarded-direction path, however, (3.36) can be integrated:

$$\boldsymbol{\psi}(s - ke_2, q, e_2) - \boldsymbol{\psi}(s - ke_1, q, e_1) = \int_{e_1}^{e_2} (\boldsymbol{b} - k\boldsymbol{a})\,\mathrm{d}e. \tag{3.48}$$

Only with suitable pathwise information could one bound a finite discrepancy, for example by $\int \|\boldsymbol{b} - k\boldsymbol{a}\|\,\mathrm{d}e$. A pooled sample residual does not by itself supply that bound.

3.6.3 How empirical derivatives enter the argument

The derivative estimates used here are local affine regressions, not direct inductance measurements. Normalize each measured current by its observed range, choose the nearest h points, and fit the two flux outputs to an intercept plus three normalized-current offsets. Squared normalized distances determine Gaussian weights. Equivalently, the least-squares design and outputs are multiplied by the square roots of those weights. The fitted slopes are divided by the physical current ranges to recover H.

Neighborhood counts $h = 40, 80, 120$ are available, with 80 used for the primary display. A neighborhood must contain at least two nominal excitation groups. Ill-conditioned affine designs and weak sensitivity vectors are excluded by numerical screens. Such rules protect a fit; they do not establish physical validity of every retained derivative. A comparison across neighborhood counts must use their common retained points, which can differ from the primary 80-neighbor population.

3.6.4 Why the different factors need not agree

The conventional factor (2.15) uses d-flux contours, arithmetic crossing averages and adjacent-group quotients. The local factor (3.43) fits two derivative components at one point. The global factor (3.46) pools derivative residuals. A reconstruction-optimal constant instead minimizes finite flux errors after a specified common table has been built. A pointwise alignment factor moves a query on an already fixed table to fit its known target flux.

These objectives differ in physical quantity, weights, population, support and numerical representation. They coincide in special exact models with appropriate sampling; agreement is not a general identity. The complete stator differential matrix, needed later for two-axis compensation and control, is

$$\boldsymbol{L}_s = \begin{bmatrix} L_{dd} & L_{dq} \\ L_{qd} & L_{qq} \end{bmatrix}. \tag{3.49}$$

The available stator outputs provide this matrix and $\boldsymbol{b}$; they do not provide an independently identified excitation flux linkage or the remaining excitation-circuit inductances.

Chapter 4

Sensitivity to Measurement and Modeling Errors

4.1 What the stationary voltage calculation assumes

The reference flux values are reconstructed from measured electrical quantities, rather than measured directly by an independent flux sensor. With the adopted dq convention,

$$\begin{aligned} u_d &= R_s i_d + \frac{\mathrm{d}\psi_d}{\mathrm{d}t} - \omega_e \psi_q, \\ u_q &= R_s i_q + \frac{\mathrm{d}\psi_q}{\mathrm{d}t} + \omega_e \psi_d. \end{aligned} \tag{4.1}$$

The signs in every subsequent sensitivity formula follow from (4.1). Here $\omega_e = p\omega_m$ is electrical angular speed in rad/s, and p is the number of pole pairs.

For stationary dq flux, set both time derivatives to zero. Moving the resistive terms and dividing by nonzero speed gives

$$\psi_d = \frac{u_q - R_s i_q}{\omega_e}, \qquad \psi_q = \frac{R_s i_d - u_d}{\omega_e}. \tag{4.2}$$

This operation is increasingly sensitive to voltage and resistance errors as $|\omega_e|$ decreases. It is not an identification method at zero speed.

If the fluxes are not stationary, the same algebra instead yields

$$\frac{u_q - R_s i_q}{\omega_e} = \psi_d + \frac{\dot{\psi}_q}{\omega_e}, \tag{4.3}$$

$$\frac{R_s i_d - u_d}{\omega_e} = \psi_q - \frac{\dot{\psi}_d}{\omega_e}. \tag{4.4}$$

Residual transients are then folded into the apparent source flux. Period averaging reduces some oscillatory components but does not prove that every operating point satisfies the stationary assumptions.

4.2 Stator-resistance errors

4.2.1 Deriving the component errors and their signs

Let the resistance used in (4.2) be $\widehat{R}_s = R_s + \Delta R_s$, while measured currents, voltage and speed are temporarily assumed exact. Define every flux error here as estimated minus true:

$$\begin{aligned}\Delta\psi_d &= \frac{u_q - (R_s + \Delta R_s)i_q}{\omega_e} - \frac{u_q - R_s i_q}{\omega_e} = -\frac{\Delta R_s}{\omega_e} i_q, \\ \Delta\psi_q &= \frac{(R_s + \Delta R_s)i_d - u_d}{\omega_e} - \frac{R_s i_d - u_d}{\omega_e} = \frac{\Delta R_s}{\omega_e} i_d.\end{aligned} \tag{4.5}$$

A positive resistance overestimate therefore tilts the d-flux field downward with increasing q-current and tilts the q-flux field upward with increasing d-current, for positive speed.

At constant speed and resistance mismatch, write $\alpha = \Delta R_s/\omega_e$. Its units are H. The apparent fields are

$$\widetilde{\psi}_d = \psi_d - \alpha i_q, \qquad \widetilde{\psi}_q = \psi_q + \alpha i_d. \tag{4.6}$$

Their stator-current differential matrix becomes

$$\widetilde{\boldsymbol{L}}_s = \begin{bmatrix} L_{dd} & L_{dq} - \alpha \\ L_{qd} + \alpha & L_{qq} \end{bmatrix}. \tag{4.7}$$

Thus a resistance mismatch can introduce an antisymmetric derivative contribution. A measured cross-derivative discrepancy alone does not identify resistance as its cause: interpolation, timing, angle conventions and other uncertainties can also contribute. If α varies with current or temperature, differentiating it produces additional terms omitted in (4.7).

4.2.2 A local d-contour ratio can remain unchanged

For constant α, the d-flux partial derivatives L_{dd} and L_{de} are unchanged. Consequently, the fixed-q ratio L_{de}/L_{dd} can remain unchanged even though the contours in the (i_d, i_q) plane tilt. At a fixed contour level, a small d-displacement compensating the tilt satisfies

$$L_{dd}\delta i_d - \alpha i_q \simeq 0, \qquad \delta i_d \simeq \frac{\alpha i_q}{L_{dd}}. \tag{4.8}$$

Different q-regions therefore move by different amounts. Crossing populations, truncation and arithmetic centroid positions can change. If the level-selection endpoints also move, the additional level perturbation enters as $L_{dd}\delta i_d \simeq \delta\lambda + \alpha i_q$.

For a fixed crossing topology with normalized weights w_m, differentiation of $\bar{d} = \sum_m w_m d_m$ gives

$$\delta\bar{d} = \sum_m w_m \delta d_m + \sum_m \delta w_m (d_m - \bar{d}), \qquad \sum_m \delta w_m = 0. \tag{4.9}$$

The second term is the contribution from changed sampling weights, not a physical trajectory of one contour point. Edge appearances, disappearances or filtering transitions can make the statistic nonsmooth, so this linearized expression has only local applicability.

4.2.3 Why joint scalar reduction can appear inconsistent

Suppose the true fluxes have an exact scalar representation $\psi_x = F_x(s, q)$ with $s = i_d + k_0 i_e$. The resistance-perturbed q-flux becomes

$$\widetilde{\psi}_q = F_q(s, q) + \alpha i_d \tag{4.10}$$

$$= F_q(s, q) + \alpha s - \alpha k_0 i_e. \tag{4.11}$$

At fixed (s, q), its final term still varies with excitation. The measured pair therefore need not satisfy the same reduction even if the underlying pair did. In sensitivity-vector notation, $\widetilde{\boldsymbol{a}} = (L_{dd}, L_{qd} + \alpha)^{\mathsf{T}}$ and $\widetilde{\boldsymbol{b}} = \boldsymbol{b}$, so

$$\widetilde{\Delta} = \Delta - \alpha L_{de}. \tag{4.12}$$

This is a concrete mathematical mechanism, not a fitted causal attribution to the data. Neither a nonzero determinant nor a tilted measured field proves which measurement error occurred.

4.3 Temperature and averaging

A local resistance approximation around temperature T_0 is

$$R_s(T) \simeq R_s(T_0)\{1 + \alpha_R(T - T_0)\}, \tag{4.13}$$

where the temperature coefficient α_R must correspond to the actual conductor and calibration convention. If a fixed resistance is used over changing temperatures, the resulting $\Delta R_s(T)$ enters (4.5). If a temperature-compensated resistance channel is used, its calibration and synchronization still matter.

Magnetic properties can depend on temperature as well. An extended quasistatic map would be $\boldsymbol{\psi}(\boldsymbol{i}, T)$, with

$$\dot{\boldsymbol{\psi}} = D_{\boldsymbol{i}}\boldsymbol{\psi}\,\dot{\boldsymbol{i}} + \frac{\partial \boldsymbol{\psi}}{\partial T}\dot{T}. \tag{4.14}$$

Slow temperature variation during different measurement groups can therefore create apparent current dependence when all records are pooled into a current-only field. It can also alter genuine incremental sensitivities. These effects cannot be removed by assuming that a recorded temperature range was a controlled, uniform magnetic temperature.

The observed stator temperature ranges in the datasets are broad, as described in Chapter 5. They document telemetry, not an isothermal experiment. In particular, an implausible rotor-temperature channel is not used to infer calibrated rotor thermal behavior.

The source records are averages of electrical quantities. In general,

$$\overline{\omega_e \psi_x} = \bar{\omega}_e\,\bar{\psi}_x + \operatorname{cov}(\omega_e, \psi_x). \tag{4.15}$$

Dividing averaged voltages by averaged speed is not exactly the same as averaging an instantaneous quotient if speed and flux fluctuate together. Repetition averaging also changes the relative weighting of acquisition records. These are possible sources of empirical-map variability, not quantified uncertainty contributions in the present evidence.

4.4 Angular errors and measurement delays

4.4.1 A constant angle offset rotates both vectors and coordinates

Let the angle used for the dq transformation exceed the true angle by δ. With rotation

$$R(\delta) = \begin{bmatrix} \cos\delta & -\sin\delta \\ \sin\delta & \cos\delta \end{bmatrix}, \tag{4.16}$$

the components formed with the shifted angle satisfy

$$\begin{bmatrix} \hat{i}_d \\ \hat{i}_q \end{bmatrix} = R(-\delta) \begin{bmatrix} i_d \\ i_q \end{bmatrix}. \tag{4.17}$$

For small δ, $\hat{i}_d \simeq i_d + \delta i_q$ and $\hat{i}_q \simeq i_q - \delta i_d$. The same component mixing applies to a consistently transformed flux vector at one physical state:

$$\hat{\psi}_d \simeq \psi_d + \delta\psi_q, \qquad \hat{\psi}_q \simeq \psi_q - \delta\psi_d. \tag{4.18}$$

However, the map arguments rotate too. A consistent transformed map is

$$\hat{\boldsymbol{\psi}}(\hat{\boldsymbol{i}}_{dq}, i_e) = R(-\delta)\boldsymbol{\psi}(R(\delta)\hat{\boldsymbol{i}}_{dq}, i_e), \tag{4.19}$$

not merely an output offset at unchanged numerical input coordinates.

A true compensation direction $(k, 0)^\mathsf{T}$ in one frame would become $R(-\delta)(k, 0)^\mathsf{T}$ in the other. An apparent q-axis compensation component can therefore reflect coordinate alignment as well as magnetic behavior. This example does not establish the presence or value of an angle error in any dataset.

4.4.2 A time delay has more than one interpretation

For a smooth scalar signal,

$$x(t - \tau) = x(t) - \tau\dot{x}(t) + O(\tau^2). \tag{4.20}$$

A delayed dq-current telemetry signal therefore differs during a dq-current transient. A delay in raw phase-current sampling before transformation can instead produce a phase error approximately $\delta \simeq \omega_e\tau$ even when the desired dq currents are steady. The correct relationship depends on where sampling, filtering and coordinate transformation occur.

If δ itself varies, the angular-speed channel can also acquire a $\dot{\delta}$ contribution. A single constant angle-offset formula does not capture this entire timing chain. Known acquisition differences motivate sensitivity questions but do not demonstrate a causal explanation of a reconstruction error or a reported oscillation.

4.5 Voltage, current and speed uncertainty

Linearize (4.2) while allowing voltage, current, resistance and speed errors. Neglect products of small errors:

$$\begin{aligned}
\Delta\psi_d &\simeq \frac{\Delta u_q - R_s\Delta i_q - i_q\Delta R_s}{\omega_e} - \psi_d\frac{\Delta\omega_e}{\omega_e}, \\
\Delta\psi_q &\simeq \frac{R_s\Delta i_d + i_d\Delta R_s - \Delta u_d}{\omega_e} - \psi_q\frac{\Delta\omega_e}{\omega_e}.
\end{aligned} \tag{4.21}$$

Voltage offsets, gain errors and speed uncertainty can therefore distort the fields differently from resistance alone. Current errors also perturb the horizontal map coordinates, an effect not exhausted by their appearance in the numerators of (4.21).

The calculation uses recorded electrical speed, not requested mechanical speed. Observed differences between actual and requested speed are consequently not themselves evidence that the wrong speed was used. Similarly, commanded inverter voltage and measured fundamental voltage are not interchangeable without a verified conversion.

4.6 Iron losses and other omitted effects

The dq voltage balance remains a balance of terminal quantities and winding flux linkages under its stated transformation and averaging assumptions. Iron loss does not automatically invalidate

that balance. It can invalidate an overly simple constitutive relation between terminal current and a single quasistatic magnetic state.

For example, a loss-aware equivalent description might separate magnetizing and loss-related currents,

$$\boldsymbol{i}_s = \boldsymbol{i}_m + \boldsymbol{i}_{\text{loss}}, \qquad \boldsymbol{\psi} = \boldsymbol{\psi}_m(\boldsymbol{i}_m, i_e, \text{magnetic state}). \tag{4.22}$$

The loss-related current can depend on frequency, flux amplitude and history. Eliminating it may introduce speed dependence, phase shifts or extra dynamics into an apparent flux map indexed only by terminal currents. This is an illustration of missing state dependence, not an identified model for the present measurements.

Replacing all such effects by a universal additive stator resistance would ignore their different directions, frequency dependences and state variables. Eddy-current dynamics, hysteresis, inverter nonlinearities and harmonic averaging can contribute in different ways. No available experiment independently isolates all of them.

4.7 What can be concluded from these sensitivities

The algebra identifies mechanisms capable of changing contour shape, derivative direction and joint reducibility. It does not identify their magnitudes from the existing current maps alone. An empirical failure of exact reduction can arise from genuine magnetic incompatibility, measurement distortion, numerical representation or a combination.

For the subsequent control discussion, source-flux errors and reduction errors both matter because the controller uses the returned flux. Their physical origins need not be identical. A speed-dependent feedforward consequence can be derived without claiming that the underlying flux error has already been uniquely explained.

Chapter 5

Experimental Investigation and Reconstruction Errors

The preceding analysis establishes what an exact reduction would require. The measurements now address a different question: how well do specific finite representations reproduce the available flux values? A measured flux field, an estimated derivative field, and a stored reduced table are different numerical objects. Agreement in one does not automatically establish agreement in the others. This chapter therefore states the target, reference map, and supported population whenever it reports an error.

5.1 Measurement populations and their limitations

Table 5.1 describes the three available datasets. Sets 1 and 2 are related measurement campaigns with the same recorded machine constants; they should not be counted as two independently characterized machines. Set 3 belongs to another machine configuration. Parameters are identified separately for each dataset.

Table 5.1: Available measurement populations and recorded electrical constants. Excitation ranges refer to nominal group labels; speed is the requested mechanical speed. Points are the averaged operating points used in the flux study.

Quantity	Set 1	Set 2	Set 3
Operating points	7,476	8,092	9,022
Excitation groups	13	14	10
Nominal excitation range (A)	2–40	0–40	3–30
Requested speed (rpm)	3,000	3,000	2,250
Pole pairs	3	3	4
Stator resistance (Ω)	0.013	0.013	0.0366185
Excitation resistance (Ω)	3.2	3.2	2.6

The nominal excitation labels are

$$\begin{aligned}
&\text{Set 1:} \quad 2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32, 36, 40 \text{ A},\\
&\text{Set 2:} \quad 0, 2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32, 36, 40 \text{ A},\\
&\text{Set 3:} \quad 3, 6, 9, 12, 15, 18, 21, 24, 27, 30 \text{ A}.
\end{aligned}$$

These labels organize acquisitions. They are not substituted for measured excitation in the transformation. A point in a nominal 17-A group retains its own measured i_e when $s = i_d + k i_e$ is evaluated. The same distinction applies to measured and requested stator currents.

The acquisition records contain electrical-period averages. Repeated records are averaged within their acquisition and operating-point groups before the retained points are pooled. In Set 3, 284 records with operating-point index zero have no assigned operating point and are excluded. The resulting population is the one in Table 5.1. Its stationary flux values are inferred from averaged voltages and measured currents using the electrical equations discussed earlier. They are empirical flux estimates, with the associated resistance, angle, speed, and stationarity assumptions.

Measured speeds span approximately 2,934–3,072 rpm in Set 1, 2,933–3,070 rpm in Set 2, and 2,247–2,252 rpm in Set 3. Recorded stator temperatures span approximately 50–80 °C in Sets 1 and 2 and 40–104 °C in Set 3. These are observed channel ranges, not evidence that all points were measured at one controlled temperature. They also do not supply a temperature correction for each point. No unique decomposition into temperature effects, magnetic behavior, sensor effects, and interpolation errors follows from these records.

5.2 From measured fields to a conventional reduced table

Figure 5.1 follows Set 2 from measured coordinates to conventional shifted coordinates. Both panels contain the same measured flux values. Only the horizontal current coordinates change in the lower panel. The visible movement is a coordinate transformation, not a correction of measured flux. If the reduction were exact, all excitation groups would agree wherever their shifted coordinates coincide. Apparent overlap is informative, but perspective and finite plotting resolution cannot establish exact agreement.

For every ψ_d field view, the horizontal X axis is i_q and the horizontal Y axis is i_d, or $s = i_d + ki_e$ after reduction. The ψ_q views use the opposite current-axis order. The same component-specific order is used in the signed-error meshes and heatmaps. This display convention changes neither the physical current definitions nor the derivative-index convention.

The shifted points are merged into a common two-coordinate field. The stored representation uses an 81×81 grid. Scattered coordinates are normalized by their ranges before triangulation; coincident coordinates receive the arithmetic mean of their flux values. Grid values are obtained by linear interpolation within retained triangles, with long-edge simplices rejected. Queries outside the supported region have no returned flux value. A subsequent lookup interpolates the stored grid linearly in both coordinates. The full operation therefore includes reduction, merging, scattered interpolation, storage, and return interpolation.

Figure 5.2 separates the merged representation from its return to physical measurement coordinates. A smooth common surface does not prove that the original groups were compatible: forming one surface has already replaced their differences by one stored answer. The returned group surfaces show how this answer varies over the original operating points.

5.3 Error definitions and coverage

For measured point j, let $\boldsymbol{\psi}_j$ be its measured flux pair and $\widehat{\boldsymbol{\psi}}_j$ the returned pair. The signed convention is

$$\boldsymbol{e}_j = \widehat{\boldsymbol{\psi}}_j - \boldsymbol{\psi}_j = \begin{bmatrix} e_{d,j} \\ e_{q,j} \end{bmatrix}. \tag{5.1}$$

On a specified supported population $\mathcal{M}$ of N points, define

$$\mathrm{RMSE}_d = \sqrt{\frac{1}{N}\sum_{j\in\mathcal{M}} e_{d,j}^2}, \qquad \mathrm{RMSE}_q = \sqrt{\frac{1}{N}\sum_{j\in\mathcal{M}} e_{q,j}^2}, \tag{5.2}$$

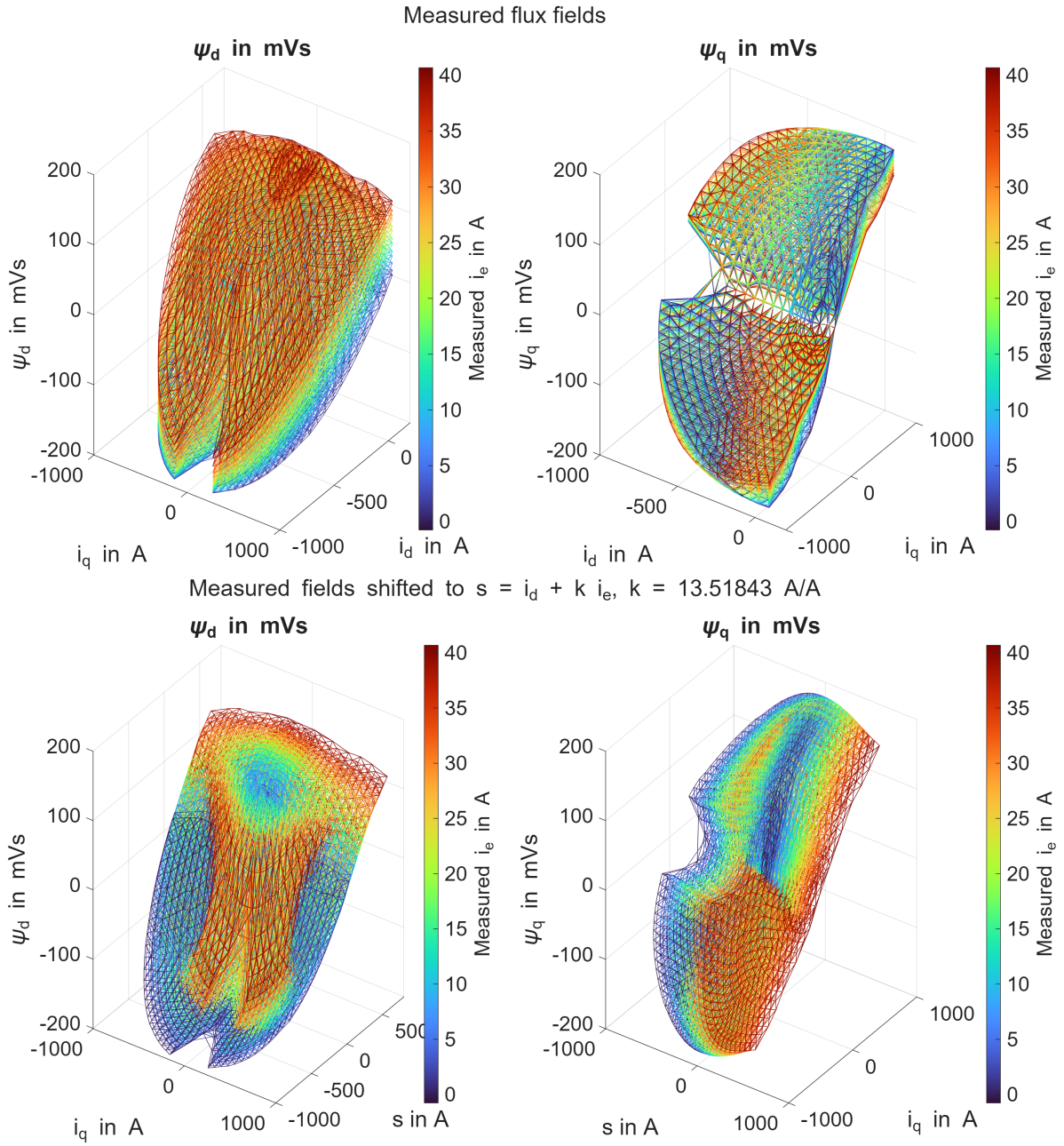


Figure 5.1: Set 2 measured flux fields before and after the conventional coordinate shift. Upper panel: measured ψ_d and ψ_q over measured stator currents. Lower panel: the same flux values after replacing i_d by $s = i_d + k_{\text{conv}} i_e$. Currents are in A, flux linkages in mVs, and color represents measured excitation in A. Original group surfaces retain their identity after shifting.

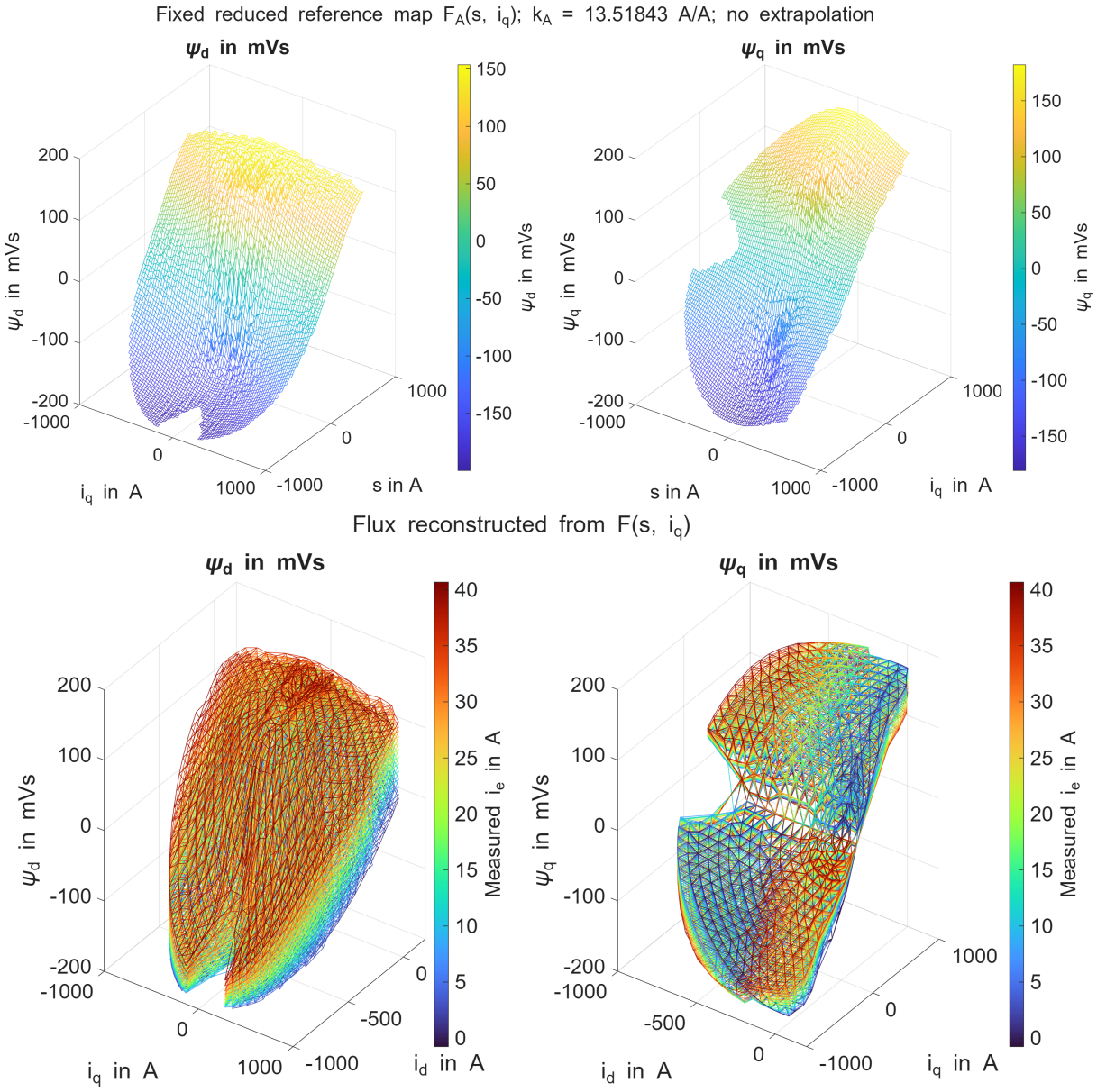


Figure 5.2: Set 2 conventional common map and returned reconstruction. Upper panel: merged flux surfaces, with (i_q, s) axes for ψ_d and (s, i_q) for ψ_q. Lower panel: returned values over physical currents, with (i_q, i_d) for ψ_d and (i_d, i_q) for ψ_q. Currents are in A and flux linkages in mVs. Missing support is not filled by extrapolation.

$$\mathrm{RMSE}_{\mathrm{joint}} = \sqrt{\frac{1}{2N}\sum_{j\in\mathcal{M}}(e_{d,j}^2 + e_{q,j}^2)} = \sqrt{\frac{\mathrm{RMSE}_d^2 + \mathrm{RMSE}_q^2}{2}}. \tag{5.3}$$

The joint statistic gives equal weight to both components. It differs by a factor $\sqrt{2}$ from the root mean squared Euclidean vector norm. Values below are in mVs; these are flux-linkage errors, not current errors.

Coverage requires an eligible population as well as a supported one:

$$C = \frac{N}{N_{\mathrm{eligible}}}. \tag{5.4}$$

An unsupported point contributes neither a zero residual nor an implicit successful prediction. A method can have an attractive conditional RMSE while failing to return values at relevant points. Fair accuracy comparisons use the same physical points and report their count. They do not become full-population comparisons merely because every number has the same unit.

Table 5.2 reports the conventional map on its own supported measurement points. These data were used to construct the map, so this is reconstruction accuracy. It is not independent prediction accuracy for an unobserved excitation group.

Table 5.2: Conventional reconstruction on each dataset's own supported points. Errors are in mVs. Coverage is relative to all operating points in that dataset.

Set	Supported / total	Coverage	RMSE d	RMSE q	Joint
1	7207 / 7476	96.40%	5.598	4.383	5.027
2	7807 / 8092	96.48%	6.249	4.476	5.435
3	8730 / 9022	96.76%	5.817	5.933	5.875

Figure 5.3 shows the signed structure behind the aggregate statistics for one Set 2 group. Positive error means the returned component exceeds the measured component. Positive and negative regions may cancel in an arithmetic mean while both increase RMSE. The nonuniform pattern also explains why one error magnitude cannot characterize every operating region. This group is illustrative; Table 5.2 pools supported points across all groups.

5.4 Contour statistics and local derivative evidence

Figure 5.4 displays the contour-level dependence of the conventional statistic. Only the shaded middle 25–75% interval determines the reported scalar factor. The full-range curves also show behavior near the limits of available contour coverage. Their variation belongs to the evaluated contour pairs; it is not a measured local function $k(i_d, i_q, i_e)$.

The lower panels measure different departures from a fixed-i_q contour-displacement argument. First, compared centroids need not share a q-coordinate. Second, evaluating the original flux surface at an arithmetic mean coordinate need not return the contour level. Table 5.3 quantifies these observations over the fitted levels. Centroid departure uses an original-group interpolant rather than the merged reduced table. Its RMS therefore measures neither reduced-map reconstruction nor missing-group prediction.

Departure can exceed reconstruction RMSE without contradiction: the quantities compare different points with different targets. Nor does a centroid departure imply an equally large factor error. The factor depends on differences between adjacent-group centroid positions; shared departures may cancel, while changing departures may bias a secant.

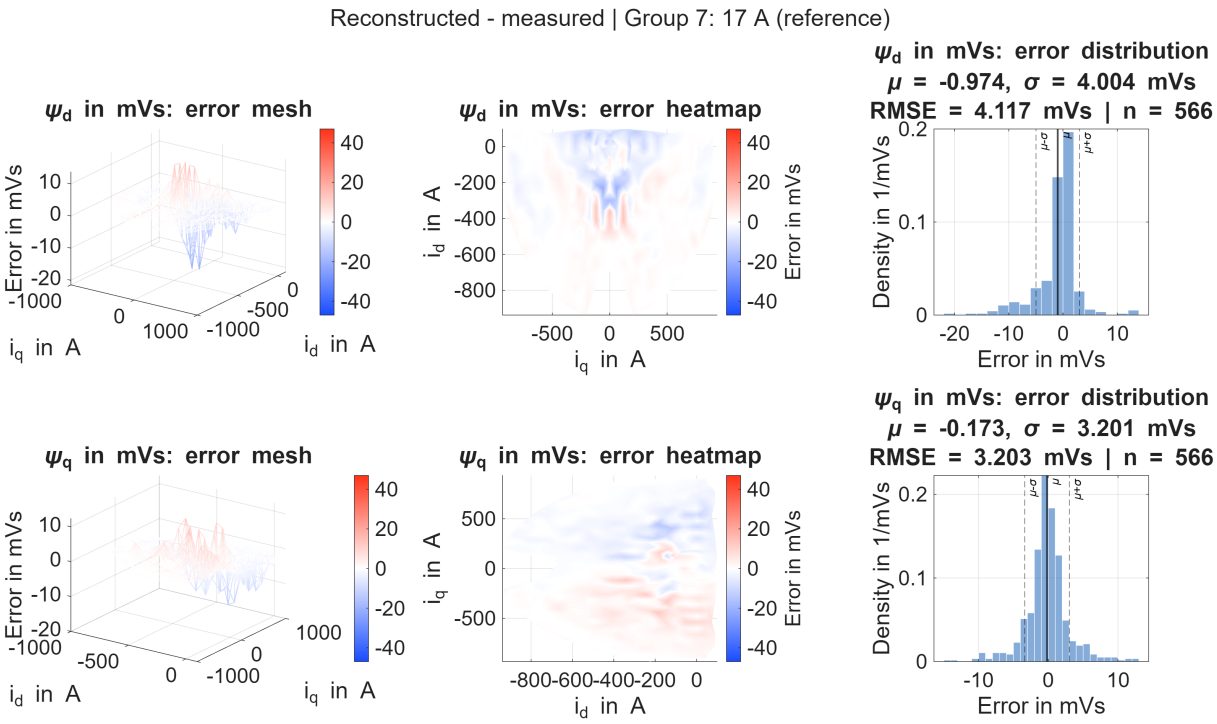


Figure 5.3: Signed conventional reconstruction errors in Set 2's nominal 17-A group, whose mean measured excitation is approximately 17.028 A. Mesh and heatmap views show returned minus measured ψ_d and ψ_q. Stator-current coordinates are in A and errors in mVs. The common color scale supports comparison of signs and component magnitudes; unsupported regions are not zero-error observations.

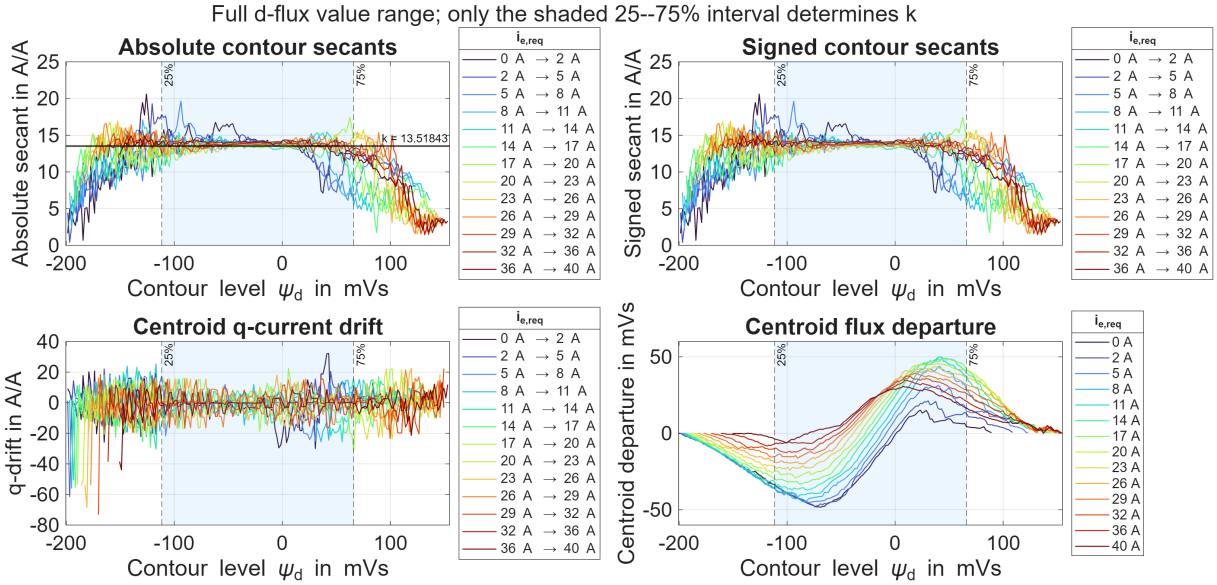


Figure 5.4: Set 2 contour diagnostics. Upper panels show absolute-numerator and signed adjacent-group secants. Lower panels show centroid q-current drift and original-group d-flux at each centroid minus its selected contour level. Flux axes are in mVs and secant or drift axes in A/A. The shaded interval identifies the levels used in the scalar estimate.

Table 5.3: Conventional factors and contour diagnostics over the fitted level interval. Departure is the RMS d-flux departure from the selected contour level. Drift is the median absolute ratio between adjacent centroid q-current and excitation differences.

Set	k_{conv} (A/A)	Departure (mVs)	Drift (A/A)	Finite secants
1	13.492116	29.223	2.547	1200
2	13.518430	28.854	3.712	1300
3	14.764191	14.176	4.706	890

An independent local calculation estimates differential sensitivities using weighted affine fits to nearby measurements in normalized three-current coordinates. The principal calculation uses 80 neighbours and requires more than one excitation group in each fit. Numerically ill-conditioned fits and insufficient d-current sensitivity are screened. This is a local regression estimate, not a derivative of the displayed reduced table.

With $\boldsymbol{a} = (L_{dd}, L_{qd})^{\mathsf{T}}$ and $\boldsymbol{b} = (L_{de}, L_{qe})^{\mathsf{T}}$, Table 5.4 compares joint least-squares factors and directional disagreement. Here

$$\eta_j = \frac{|a_{d,j} b_{q,j} - a_{q,j} b_{d,j}|}{\|\boldsymbol{a}_j\| \, \|\boldsymbol{b}_j\|}, \qquad \rho(k) = \frac{\sqrt{\sum_j \|\boldsymbol{b}_j - k\boldsymbol{a}_j\|^2}}{\sqrt{\sum_j \|\boldsymbol{b}_j\|^2}}. \tag{5.5}$$

For the local column of ρ, the factor varies with j. These residuals are dimensionless ratios of inductance norms. They are not percentages of flux reconstruction error.

Table 5.4: Primary 80-neighbour derivative diagnostics. Retained populations contain 7476, 8092, and 9022 points. Factors are in A/A; η and ρ are dimensionless.

Set	Global LS	Median local LS	Median η	$\rho(k_{\mathrm{conv}})$	$\rho(k_{\mathrm{der}})$	$\rho(k_{\mathrm{loc}})$
1	13.8374	13.1790	0.0898	0.1619	0.1600	0.0883
2	13.9514	13.2134	0.0871	0.1562	0.1532	0.0831
3	14.4482	14.3043	0.1442	0.1973	0.1962	0.1379

Changing the neighbourhood changes numerical slopes and sometimes their supported population. A neighbourhood comparison must itself use a common mask. Nonzero residuals support imperfect common scalar compensation in the estimated sensitivities. They do not uniquely identify its physical cause or provide uncertainty bounds for the underlying differential inductances.

5.5 Pointwise scalar alignment as a diagnostic reference

How much error remains if every measurement receives its own scalar factor? Fix the conventional reduced map $\boldsymbol{F}_A$ and the measurement's i_q. For $i_{e,j} \neq 0$, consider

$$k_j^\star \in \arg\min_k \|\boldsymbol{F}_A(i_{d,j} + k i_{e,j}, i_{q,j}) - \boldsymbol{\psi}_j\|^2 , \tag{5.6}$$

restricted to supported table coordinates. The known target flux participates directly in identification. This is an alignment diagnostic on available data, not a predictor that infers $k_j^\star$ from currents alone.

As k varies, the returned flux pair traces a one-dimensional curve in the two-dimensional (ψ_d, ψ_q) plane. An arbitrary target pair need not lie on this curve. Even the best scalar factor can leave a nonzero residual in either component. Minimizing their joint squared error does not force either component individually to be exact.

The finite bilinear table makes this geometry explicit. At fixed i_q, every supported d-coordinate grid interval maps to a straight flux segment. Let its endpoints be $\boldsymbol{f}_0$ and $\boldsymbol{f}_1$, put $\boldsymbol{v} = \boldsymbol{f}_1 - \boldsymbol{f}_0$, and parameterize it by

$$\boldsymbol{f}(t) = \boldsymbol{f}_0 + t\boldsymbol{v}, \qquad 0 \le t \le 1. \tag{5.7}$$

For a nonflat segment, differentiating its squared target distance gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \|\boldsymbol{f}_0 + t\boldsymbol{v} - \boldsymbol{\psi}_j\|^2 = 2\boldsymbol{v}^\mathsf{T}(\boldsymbol{f}_0 + t\boldsymbol{v} - \boldsymbol{\psi}_j), \tag{5.8}$$

$$t^\star = \mathrm{clip}_{[0,1]} \left(\frac{\boldsymbol{v}^\mathsf{T}(\boldsymbol{\psi}_j - \boldsymbol{f}_0)}{\boldsymbol{v}^\mathsf{T}\boldsymbol{v}} \right). \tag{5.9}$$

Comparing this candidate over every supported segment finds the minimum on the represented curve. The winning reduced coordinate $s_j^\star$ gives

$$k_j^\star = \frac{s_j^\star - i_{d,j}}{i_{e,j}}. \tag{5.10}$$

A flat minimizing segment or separated equally good coordinates makes the factor nonunique. At zero excitation the factor cannot move the query; near zero its recovery is sensitive to small coordinate changes. Low excitation, weak sensitivity, missing support, and nonunique solutions are screened. These restrictions affect coverage and must accompany any reported error reduction.

5.6 Pointwise two-axis alignment and identifiability

Allowing two factors changes the attainable flux set. Fix a reference map $\boldsymbol{F}_{\mathrm{ref}}$ and write

$$\boldsymbol{s}_j = \begin{bmatrix} i_{d,j} \\ i_{q,j} \end{bmatrix} + i_{e,j} \begin{bmatrix} k_{d,j} \\ k_{q,j} \end{bmatrix}, \qquad \min_{k_{d,j}, k_{q,j}} \|\boldsymbol{F}_{\mathrm{ref}}(\boldsymbol{s}_j) - \boldsymbol{\psi}_j\|^2 . \tag{5.11}$$

This fixed reference is constructed with the separately identified constant pairs, in A/A,

$$(k_{d,\mathrm{ref}}, k_{q,\mathrm{ref}}) = \begin{cases} (14.336652, -0.025602), & \text{Set 1}, \\ (14.186955, -0.091824), & \text{Set 2}, \\ (14.627088, -0.787874), & \text{Set 3}. \end{cases} \tag{5.12}$$

These pairs define the reference; they are not the subsequently refined consistent-global solution.

For nonzero excitation, any supported reference coordinate pair is algebraically attainable by a factor pair. The flux target still need not belong to the image of the supported table. Within one cell that image is described by

$$\boldsymbol{F}(u, v) = \boldsymbol{a} + \boldsymbol{b}u + \boldsymbol{c}v + \boldsymbol{d}uv, \qquad (u, v) \in [0, 1]^2. \tag{5.13}$$

Here the bold coefficients describe one bilinear cell, rather than the differential sensitivity vectors used above. Exact interior matches solve two equations for u, v. The coordinate-to-flux Jacobian has columns $\boldsymbol{b} + \boldsymbol{d}v$ and $\boldsymbol{c} + \boldsymbol{d}u$. Nonsingularity permits a locally unique inverse. It does not rule out another inverse elsewhere. Singular cells, multiple branches, and targets outside the image require separate treatment. Minimum-distance candidates include cell boundaries and singular loci as well as exact interior roots.

The diagnostic excludes low excitation, separated minimizing branches, and poorly conditioned recovered inverses. Geometric feasibility of choosing some supported coordinates holds for every measurement in this two-axis calculation, but differs from recovering an identifiable factor pair.

Low-excitation exclusions number 280, 784, and 0; multiple-solution flags number 1566, 1602, and 2925; ill-conditioning flags number 11, 11, and 25. In these datasets the category counts account for all exclusions. No numerical failure is recorded, and retained two-axis populations contain 5619, 5695, and 6072 points. An excluded factor is not evidence of a large minimum flux distance: a target can have accurate but nonunique inverse coordinates.

5.7 Comparing diagnostic references on identical physical points

Table 5.5 uses the intersection of four valid populations. Conventional and scalar pointwise rows use $\boldsymbol{F}_A$. The two-axis pointwise row uses the distinct fixed $\boldsymbol{F}_{\mathrm{ref}}$. The consistent-global row uses its own matching table, rebuilt while identifying its constant pair. Equal physical populations remove one source of unfair comparison, but do not make the reference maps or fitting freedoms identical.

Table 5.5: Reconstruction and alignment diagnostics on one shared physical population per dataset. Errors are in mVs. Shared counts are 5181/7476, 5275/8092, and 5697/9022 for Sets 1–3. Pointwise rows use measured target fluxes to identify each point's factors.

Set	Representation or diagnostic	RMSE d	RMSE q	Joint
1	Conventional constant	5.59901	4.13100	4.92007
	Consistent global two-axis	4.89164	4.17327	4.54667
	Pointwise scalar, fixed $\boldsymbol{F}_A$	1.67422	3.57731	2.79286
	Pointwise two-axis, fixed $\boldsymbol{F}_{\mathrm{ref}}$	0.06832	0.06301	0.06571
2	Conventional constant	6.23885	4.28485	5.35178
	Consistent global two-axis	5.17633	4.15420	4.69317
	Pointwise scalar, fixed $\boldsymbol{F}_A$	1.71964	3.69563	2.88226
	Pointwise two-axis, fixed $\boldsymbol{F}_{\mathrm{ref}}$	0.01045	0.01047	0.01046
3	Conventional constant	5.18494	5.55793	5.37467
	Consistent global two-axis	5.15158	5.11938	5.13551
	Pointwise scalar, fixed $\boldsymbol{F}_A$	2.23517	4.21693	3.37479
	Pointwise two-axis, fixed $\boldsymbol{F}_{\mathrm{ref}}$	0.19942	0.27450	0.23991

The scalar diagnostic substantially reduces d-component error but retains appreciable joint error. This agrees with the curve geometry: one coordinate cannot generally place an arbitrary two-component target on its image. The much smaller two-axis errors demonstrate additional local alignment freedom. They do not establish an exact global reduction, a unique physical transformation, or an implementable online parameter function. Targets were already known, and some favorable alignments were excluded because their factors were not identifiable.

The distinctions can now be stated precisely. Centroid departure tests where a contour average lies on its original surface. Derivative residual tests local directional compensation. Reconstruction tests a map built from available data. Pointwise alignment tests how closely a fixed reference approaches known targets with individually chosen coordinates. Whole-group prediction instead removes a nominal excitation group before identifying a representation and querying that group's flux. Only the last experiment directly addresses unseen-group prediction, and even it remains conditional on support. Chapter 7 uses the existing prediction evidence to compare practical representations; none of the diagnostic improvements above substitutes for it.

Chapter 6

Implications for Current Control

The practical question is not whether a reduced flux map reproduces every measured flux value exactly. It is how the remaining error enters the voltage command, how feedback responds, and when the available voltage prevents that response. Answering this question requires three distinct objects: the physical machine, the mathematical model used for controller design, and the particular controller structure. Their equations can resemble one another while their assumptions differ substantially.

The available controller implementations provide concrete examples of PI feedback, map-based feedforward, and different treatments of voltage limits. They do not establish which parameter settings and operating modes were active during the three measurement campaigns. Consequently, the implementation properties discussed below are verified properties of available structures, whereas their activation in an individual experiment remains unverified. No measured closed-loop performance is inferred from the reconstruction errors alone.

6.1 From flux maps to dynamic machine equations

Let $\boldsymbol{i}_s = (i_d, i_q)^\mathsf{T}$, $\boldsymbol{u} = (u_d, u_q)^\mathsf{T}$, and introduce the rotation matrix

$$\boldsymbol{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \tag{6.1}$$

Its action is a quarter turn: $\boldsymbol{J}\boldsymbol{\psi} = (-\psi_q, \psi_d)^\mathsf{T}$. With the adopted electrical-angle convention, the voltage equations become

$$\boldsymbol{u} = R_s \boldsymbol{i}_s + \frac{\mathrm{d}\boldsymbol{\psi}}{\mathrm{d}t} + \omega_e \boldsymbol{J}\boldsymbol{\psi}. \tag{6.2}$$

The three terms describe resistive voltage, the voltage needed to change flux, and the voltage associated with rotation of the flux vector. The rotational term exists even when all three currents are constant.

For a differentiable quasistatic flux map $\boldsymbol{\psi}(i_d, i_q, i_e)$, the chain rule gives

$$\dot{\psi}_d = L_{dd}\dot{i}_d + L_{dq}\dot{i}_q + L_{de}\dot{i}_e, \tag{6.3}$$

$$\dot{\psi}_q = L_{qd}\dot{i}_d + L_{qq}\dot{i}_q + L_{qe}\dot{i}_e. \tag{6.4}$$

Here every derivative is evaluated at the instantaneous operating point. In matrix form, define

$$\boldsymbol{L}_s = \begin{bmatrix} L_{dd} & L_{dq} \\ L_{qd} & L_{qq} \end{bmatrix}, \qquad \boldsymbol{\ell}_e = \begin{bmatrix} L_{de} \\ L_{qe} \end{bmatrix}. \tag{6.5}$$

Substitution into the voltage equations yields

$$u_d = R_s i_d + L_{dd}\dot{i}_d + L_{dq}\dot{i}_q + L_{de}\dot{i}_e - \omega_e \psi_q, \tag{6.6}$$

$$u_q = R_s i_q + L_{qd}\dot{i}_d + L_{qq}\dot{i}_q + L_{qe}\dot{i}_e + \omega_e \psi_d. \tag{6.7}$$

Thus a changing excitation current disturbs both stator axes, including at zero speed. A controller that compensates only rotational voltage does not cancel this excitation-induced transient.

The differential inductances in these equations must not be replaced indiscriminately by secant inductances. For example, if $\psi_d = L_0 i_d + c i_d^3$ along a fixed slice, then $\psi_d/i_d = L_0 + c i_d^2$, whereas $\partial\psi_d/\partial i_d = L_0 + 3c i_d^2$. The first ratio describes a finite relation to an arbitrarily chosen origin; the derivative describes the incremental voltage required to change current. With excitation present, ψ_d/i_d can additionally include an excitation offset and becomes singular at $i_d = 0$ even when the differential inductance is regular.

If $\det \boldsymbol{L}_s \neq 0$, the nonlinear stator-current dynamics can be written exactly within this quasistatic map model as

$$\dot{\boldsymbol{i}}_s = \boldsymbol{L}_s^{-1}\left(\boldsymbol{u} - R_s \boldsymbol{i}_s - \omega_e \boldsymbol{J}\boldsymbol{\psi} - \boldsymbol{\ell}_e \dot{i}_e\right). \tag{6.8}$$

This is not yet a linear control plant: both the inverse matrix and the flux vector depend on the currents. Moreover, mathematical invertibility alone is insufficient for robust numerical inversion. A nearly singular matrix amplifies flux-derivative errors and voltage uncertainty. Cross-saturation matters through the determinant $L_{dd}L_{qq} - L_{dq}L_{qd}$ and through the directions of its singular vectors, not merely through the size of either diagonal entry.

The model assumes that the flux map captures the relevant magnetic state. Hysteresis, eddy-current dynamics, and temperature-dependent changes require additional states or dependencies if they are significant. Reciprocity, $L_{dq} = L_{qd}$, follows from an appropriate smooth magnetic coenergy model under its assumptions; it should not be imposed as an experimentally established identity on independently interpolated noisy maps.

6.2 The local current-control plant

To obtain a tractable linear model, consider a steady operating point $(\boldsymbol{i}_{s0}, i_{e0}, \omega_{e0})$. Use δ for a small perturbation about that point, and freeze the differential inductances at their local values. To first order,

$$\delta\boldsymbol{\psi} = \boldsymbol{L}_0 \delta\boldsymbol{i}_s + \boldsymbol{\ell}_{e0}\delta i_e. \tag{6.9}$$

Because the reference operating point is steady, differentiation gives $\delta\dot{\boldsymbol{\psi}} = \boldsymbol{L}_0 \delta\dot{\boldsymbol{i}}_s + \boldsymbol{\ell}_{e0}\delta\dot{i}_e$. Linearizing the rotational product also gives

$$\delta(\omega_e \boldsymbol{J}\boldsymbol{\psi}) = \omega_{e0}\boldsymbol{J}\delta\boldsymbol{\psi} + \boldsymbol{J}\boldsymbol{\psi}_0 \delta\omega_e. \tag{6.10}$$

Neglecting resistance variation within this local calculation, substitution produces

$$\begin{aligned}\boldsymbol{L}_0 \delta\dot{\boldsymbol{i}}_s + (R_s\boldsymbol{I} + \omega_{e0}\boldsymbol{J}\boldsymbol{L}_0)\delta\boldsymbol{i}_s = \delta\boldsymbol{u} &- \boldsymbol{\ell}_{e0}\delta\dot{i}_e \\ &- \omega_{e0}\boldsymbol{J}\boldsymbol{\ell}_{e0}\delta i_e - \boldsymbol{J}\boldsymbol{\psi}_0\delta\omega_e.\end{aligned} \tag{6.11}$$

An excitation change therefore has both a derivative contribution and a speed-dependent contribution. A speed change couples through the existing flux even if the currents have not yet changed.

Within transfer functions in this chapter, s denotes the Laplace variable, not the reduced current coordinate of earlier chapters. With zero initial perturbations, Laplace transformation gives

$$\begin{aligned}\boldsymbol{M}(s)\delta\boldsymbol{i}_s(s) = \delta\boldsymbol{u}(s) - (s\boldsymbol{I} + \omega_{e0}\boldsymbol{J})\boldsymbol{\ell}_{e0}\delta i_e(s) \\ - \boldsymbol{J}\boldsymbol{\psi}_0\delta\omega_e(s),\end{aligned} \tag{6.12}$$

$$\boldsymbol{M}(s) = s\boldsymbol{L}_0 + R_s\boldsymbol{I} + \omega_{e0}\boldsymbol{J}\boldsymbol{L}_0. \tag{6.13}$$

Writing $\boldsymbol{M} = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ makes the cross-coupling explicit:

$$a = R_s + sL_{dd} - \omega_{e0}L_{qd}, \qquad b = sL_{dq} - \omega_{e0}L_{qq}, \tag{6.14}$$

$$c = sL_{qd} + \omega_{e0}L_{dd}, \qquad d = R_s + sL_{qq} + \omega_{e0}L_{dq}. \tag{6.15}$$

Where its determinant is nonzero, the voltage-to-current transfer matrix is

$$\boldsymbol{G}(s) = \boldsymbol{M}(s)^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}. \tag{6.16}$$

This expression shows why two independently tuned scalar loops are an approximation: a voltage perturbation on one axis generally produces current changes on both axes.

At standstill with negligible cross derivatives, or after ideal cancellation of the relevant couplings, one axis reduces to

$$L\dot{i} + R_s i = v, \qquad G(s) = \frac{1}{Ls + R_s} = \frac{1/R_s}{1 + sL/R_s}. \tag{6.17}$$

The time constant $T_e = L/R_s$ then has a direct physical meaning: inductance resists current change, while resistance dissipates magnetic energy. For a coupled passive local model at zero speed, the corresponding modes depend on the eigenvalues of $\boldsymbol{L}_0$, rather than separately on its diagonal entries.

There is an additional distinction between measured-state decoupling and reference-based feedforward. If $\omega_e\boldsymbol{J}\boldsymbol{\psi}(\boldsymbol{i}_s, i_e)$ is evaluated at the actual state and canceled exactly, its local state-dependent contribution disappears from the plant seen by feedback. If the map is evaluated at current references, it predicts the voltage of the desired operating point. During tracking errors, the actual rotational voltage still depends on the actual currents. The scalar RL model is therefore a design approximation in that situation, not an exact consequence of adding a feedforward term.

The frozen linearization is also local in time. Along a changing operating trajectory, variations of $\boldsymbol{L}_s$ multiply the nonzero nominal current derivative. Those additional terms cannot be discarded merely because the current perturbation itself is small.

6.3 What a PI regulator does

For one axis define $e = i^\star - i$, where $i^\star$ is the current reference. An ideal continuous PI regulator has

$$u_{\mathrm{PI}} = K_p e + x, \qquad \dot{x} = K_i e, \qquad C(s) = K_p + \frac{K_i}{s}. \tag{6.18}$$

The proportional term reacts immediately to the present error. The integral state accumulates error over time and can maintain a nonzero compensating voltage after the current error has vanished. Its units are voltage; K_p has units of voltage per current and K_i of voltage per current per second.

To isolate these effects, let w be an additive voltage disturbance on the right-hand side of the RL equation, so that $i = G(C(i^\star - i) + w)$. Collecting the terms involving i gives

$$(1 + GC)i = GCi^\star + Gw, \tag{6.19}$$

and hence

$$i = \frac{GC}{1+GC} i^\star + \frac{G}{1+GC} w. \tag{6.20}$$

For the RL plant, the two transfer functions are

$$\frac{i}{i^\star} = \frac{K_p s + K_i}{Ls^2 + (R_s + K_p)s + K_i}, \tag{6.21}$$

$$\frac{i}{w} = \frac{s}{Ls^2 + (R_s + K_p)s + K_i}. \tag{6.22}$$

At zero frequency the first tends to one and the second to zero. For a stable loop with sufficient voltage, a constant reference and a constant voltage disturbance can therefore be tracked and rejected with zero steady current error. In that equilibrium the integral state supplies the missing voltage.

The conclusion is conditional. The final-value argument requires stability; saturation can prevent the commanded voltage from being applied; sensor bias changes the meaning of zero measured error; and time-varying disturbances contain frequencies at which rejection is incomplete. A PI regulator does not erase every error instantly. Increasing its gains indefinitely is also impossible because computation, sampling, modulation, and measurement introduce phase lag.

For the coupled plant, the same algebra must preserve matrix order. With a diagonal or full controller $\boldsymbol{C}(s)$,

$$\delta \boldsymbol{i}_s = (\boldsymbol{I} + \boldsymbol{GC})^{-1} \boldsymbol{GC}\, \delta \boldsymbol{i}_s^\star + (\boldsymbol{I} + \boldsymbol{GC})^{-1} \boldsymbol{G} \boldsymbol{w}. \tag{6.23}$$

This is the appropriate starting point when cross-coupling remains significant. Scalar formulas are useful for understanding each mechanism, but they do not prove stability of the full nonlinear saturated system.

6.3.1 An ideal magnitude-optimum derivation

An instructive design model appends one small first-order lag to the RL plant:

$$G_a(s) = \frac{1}{(Ls + R_s)(1 + \tau_\Sigma s)}. \tag{6.24}$$

The parameter $\tau_\Sigma > 0$ represents an aggregated small lag. Approximating an actual pure delay by this first-order element is justified only over a restricted frequency range. It does not give an exact sampled-data representation of PWM and computation.

Choose the PI zero to cancel the electrical pole:

$$T_i = \frac{K_p}{K_i} = \frac{L}{R_s}, \qquad C(s) = K_p \frac{s + R_s/L}{s}. \tag{6.25}$$

After cancellation,

$$C(s) G_a(s) = \frac{K_p/L}{s(1 + \tau_\Sigma s)}. \tag{6.26}$$

Set $a = K_p/L$. The closed reference response becomes

$$T(s) = \frac{a}{\tau_\Sigma s^2 + s + a}. \tag{6.27}$$

To see the magnitude criterion rather than simply quote its gains, evaluate the inverse squared magnitude at angular frequency Ω:

$$\begin{aligned} |T(\mathrm{j}\Omega)|^{-2} &= \left(1 - \frac{\tau_\Sigma}{a}\Omega^2\right)^2 + \frac{\Omega^2}{a^2} \\ &= 1 + \left(\frac{1}{a^2} - \frac{2\tau_\Sigma}{a}\right)\Omega^2 + \frac{\tau_\Sigma^2}{a^2}\Omega^4. \end{aligned} \tag{6.28}$$

Canceling the quadratic term makes the response maximally flat through this order near zero frequency. Therefore $1 - 2\tau_\Sigma a = 0$, giving

$$K_p = \frac{L}{2\tau_\Sigma}, \qquad K_i = \frac{R_s}{2\tau_\Sigma}, \qquad T(s) = \frac{1}{1 + 2\tau_\Sigma s + 2\tau_\Sigma^2 s^2}. \tag{6.29}$$

The resulting second-order damping ratio is $1/\sqrt{2}$. This calculation explains both the pole-zero cancellation and the role of delay in limiting gain. It assumes known positive L and R_s, effective decoupling, unity actuator gain, a single small lag, and no saturation. Neither the optimum nor its stability guarantees automatically survive violations of those assumptions.

6.3.2 Verified structures and the limits of their interpretation

One available synchronous-machine controller uses local diagonal inductance values and a positive configurable dynamics factor γ. Before additional multiplicative gain adjustments, its gain calculation is

$$K_{p,d} = \frac{L_{dd}}{8\gamma T_{\mathrm{PWM}}}, \quad K_{p,q} = \frac{L_{qq}}{8\gamma T_{\mathrm{PWM}}}, \quad K_{i,d} = K_{i,q} = \frac{R_s}{4\gamma T_{\mathrm{PWM}}}. \tag{6.30}$$

Consequently, its nominal ratios are

$$\frac{K_{p,d}}{K_{i,d}} = \frac{L_{dd}}{2R_s}, \qquad \frac{K_{p,q}}{K_{i,q}} = \frac{L_{qq}}{2R_s}. \tag{6.31}$$

They differ from the electrical time constants used for exact cancellation in the preceding derivation. Independent gain adjustments can change these ratios further. It would therefore be incorrect to identify this synchronous branch with the ideal magnitude-optimum calculation solely because it contains inductance, resistance, and a switching period. A separate machine-type branch uses a delay estimate of $2.5T_{\mathrm{PWM}}$ and a different gain formula; that branch does not establish the synchronous-machine tuning.

A second, simpler available structure uses $K_{p,d} = L_{dd}/(2T_{\mathrm{PWM}})$ and $K_{i,d} = R_s/(2T_{\mathrm{PWM}})$, with analogous q-axis gains. Its nominal gain ratio does equal L_{dd}/R_s. Its discrete integral update adds $K_i T_{\mathrm{PWM}} e$ and its output adds proportional, integral, and rotational feedforward voltages. The inspected control path does not contain the voltage allocator and back-calculation path described below. This simpler structure illustrates why features from different available implementations must not be combined into an imaginary single controller. Its gain formula alone also does not verify an aggregate physical delay equal to the switching period.

6.4 Flux feedforward and the meaning of a flux error

Feedforward predicts a voltage before an error has accumulated. For a desired current operating point, a stationary prediction is

$$\boldsymbol{u}_{\mathrm{ff}} = \widehat{R}_s \boldsymbol{i}_s^\star + \omega_e \boldsymbol{J} \widehat{\boldsymbol{\psi}}(\boldsymbol{i}_s^\star, i_e^\star). \tag{6.32}$$

The richer inspected controller includes these resistive and rotational terms, together with switchable differential feedforward. Its normal map inputs are requested d-axis, q-axis, and excitation currents. An emulator-related selection substitutes internal measured d/q currents for the stator references. The excitation coordinate remains a requested current in the inspected path. Thus the map arguments are mode-dependent.

The switchable differential terms use diagonal inductances and finite differences of lookup coordinates. In the reduced d coordinate this includes the change of $i_d^\star + k i_e^\star$. Such a calculation

is a particular approximation to the flux derivative; it is not a verified implementation of every cross derivative in Equation (6.7). The simpler structure described above adds only rotational flux feedforward in its inspected control path.

Define the model error at identical current arguments by

$$\Delta\boldsymbol{\psi}(\boldsymbol{i}_s, i_e) = \widehat{\boldsymbol{\psi}}(\boldsymbol{i}_s, i_e) - \boldsymbol{\psi}(\boldsymbol{i}_s, i_e). \tag{6.33}$$

This sign convention is essential: $\Delta\boldsymbol{\psi}$ is model minus true flux. For rotational feedforward at the same electrical speed,

$$\Delta\boldsymbol{u}_{\mathrm{rot}} = \omega_e \boldsymbol{J}\Delta\boldsymbol{\psi}, \qquad \Delta u_d = -\omega_e \Delta\psi_q, \quad \Delta u_q = \omega_e \Delta\psi_d. \tag{6.34}$$

Because $\boldsymbol{J}$ preserves Euclidean length,

$$\|\Delta\boldsymbol{u}_{\mathrm{rot}}\| = |\omega_e| \, \|\Delta\boldsymbol{\psi}\| . \tag{6.35}$$

A flux error measured in volt-seconds therefore produces a voltage error proportional to electrical speed. This is a statement about feedforward voltage, not a direct equality between flux error and current error.

If speed is also estimated incorrectly, with $\widehat{\omega}_e = \omega_e + \Delta\omega_e$, then

$$\widehat{\omega}_e \boldsymbol{J}\widehat{\boldsymbol{\psi}} - \omega_e \boldsymbol{J}\boldsymbol{\psi} = \omega_e \boldsymbol{J}\Delta\boldsymbol{\psi} + \Delta\omega_e \boldsymbol{J}\boldsymbol{\psi} + \Delta\omega_e \boldsymbol{J}\Delta\boldsymbol{\psi}. \tag{6.36}$$

The first-order approximation drops only the final product. A large true flux can therefore make even a modest speed error relevant independently of map reconstruction error.

For a hypothetical full trajectory feedforward that also includes $\mathrm{d}\widehat{\boldsymbol{\psi}}/\mathrm{d}t$, compare modeled and true voltages along the same current trajectory. At identical speed,

$$\Delta\boldsymbol{u}_{\mathrm{full}} = \Delta R_s \boldsymbol{i}_s + \frac{\mathrm{d}\Delta\boldsymbol{\psi}}{\mathrm{d}t} + \omega_e \boldsymbol{J}\Delta\boldsymbol{\psi}. \tag{6.37}$$

Componentwise,

$$\Delta u_d = \Delta R_s i_d + \dot{\Delta\psi}_d - \omega_e \Delta\psi_q, \tag{6.38}$$

$$\Delta u_q = \Delta R_s i_q + \dot{\Delta\psi}_q + \omega_e \Delta\psi_d. \tag{6.39}$$

This derivative term belongs to the comparison of full dynamic voltage predictions. It must not be attributed to a controller that implements only rotational feedforward. In such a controller, unmodeled flux-derivative effects remain part of the plant and the voltage demand to be supplied by feedback.

If both maps are smooth, the derivative of their difference is

$$\dot{\Delta\boldsymbol{\psi}} = \Delta\boldsymbol{L}_s \dot{\boldsymbol{i}}_s + \Delta\boldsymbol{\ell}_e \dot{i}_e. \tag{6.40}$$

A small pointwise flux error does not necessarily imply small derivative error. Interpolation can reproduce values closely while changing slopes, particularly near sparse support or cell boundaries. Conversely, a constant flux offset has zero derivative error but still produces a speed-dependent rotational voltage error.

When feedforward uses reference currents, its total mismatch also contains tracking mismatch:

$$\begin{aligned}\widehat{\boldsymbol{\psi}}(\boldsymbol{i}_s^\star, i_e^\star) - \boldsymbol{\psi}(\boldsymbol{i}_s, i_e) = {} & \Delta\boldsymbol{\psi}(\boldsymbol{i}_s^\star, i_e^\star) \\ & + \boldsymbol{\psi}(\boldsymbol{i}_s^\star, i_e^\star) - \boldsymbol{\psi}(\boldsymbol{i}_s, i_e).\end{aligned} \tag{6.41}$$

Only the first term is the map error at common arguments. To first order the second is $\boldsymbol{L}_s(\boldsymbol{i}_s^\star - \boldsymbol{i}_s) + \boldsymbol{\ell}_e(i_e^\star - i_e)$. This distinction prevents ordinary tracking transients, including excitation-current lag, from being misidentified as reconstruction error.

6.4.1 Connecting coordinate-factor error to voltage sensitivity

To isolate the effect of an erroneous scalar factor, consider a deliberately restricted comparison: keep a differentiable reduced map $\boldsymbol{F}$ fixed and change only its coordinate factor from k_0 to $k_0 + \delta k$. At the same physical currents, let $s_0 = i_d + k_0 i_e$. Taylor expansion in the coordinate displacement gives

$$\boldsymbol{F}(s_0 + \delta k\, i_e, i_q) - \boldsymbol{F}(s_0, i_q) = \boldsymbol{F}_s(s_0, i_q)\, i_e \delta k + O((i_e \delta k)^2). \tag{6.42}$$

The vector $\boldsymbol{F}_s$ contains the sensitivities of both flux components to the reduced d coordinate. At first order, rotational voltage sensitivity follows immediately:

$$\Delta \boldsymbol{u}_{\mathrm{rot}} \simeq \omega_e \boldsymbol{J} \boldsymbol{F}_s\, i_e \delta k. \tag{6.43}$$

Thus speed, excitation current, local map slope, and factor error enter together. A single percentage difference between two factors does not determine the voltage consequence across an operating region.

For constant k_0 and constant δk, differentiation of the first-order flux error along a trajectory gives

$$\frac{\mathrm{d}\Delta\boldsymbol{\psi}}{\mathrm{d}t} \simeq \delta k \left[\boldsymbol{F}_s \dot{i}_e + i_e \left(\boldsymbol{F}_{ss}(\dot{i}_d + k_0 \dot{i}_e) + \boldsymbol{F}_{sq} \dot{i}_q \right) \right]. \tag{6.44}$$

Even a constant factor error can therefore produce a time-varying voltage mismatch when the currents change. The first term concerns excitation-current change directly; the others concern changing sensitivity along the map. This is a local smooth-map explanation, not a calculation of the actual experimental transient.

The comparison deliberately holds $\boldsymbol{F}$ fixed. Rebuilding a common reduced map after changing k also changes its fitted values, support, and interpolation geometry. The resulting reconstruction difference cannot generally be attributed to Equation (6.42) alone. Nor does this argument assume that the true machine possesses an exact scalar reduction: any irreducible map discrepancy remains an additional error. This distinction connects the diagnostic factor studies to control sensitivity without treating a fitted factor as an independently known physical constant.

6.5 How feedback responds to imperfect feedforward

Suppose that the ideal stationary feedforward is $\boldsymbol{u}_0$ and the model supplies $\boldsymbol{u}_0 + \Delta\boldsymbol{u}_{\mathrm{ff}}$. With sufficient voltage and a stable PI loop, a possible steady equilibrium is

$$\boldsymbol{e} = \boldsymbol{0}, \qquad \boldsymbol{x} = -\Delta\boldsymbol{u}_{\mathrm{ff}}. \tag{6.45}$$

Thus nonzero feedforward error can coexist with accurate current tracking. The controller must build the compensating integral voltage during the approach to equilibrium, and that approach can include current deviations.

For the scalar unsaturated RL approximation with a constant reference, Equation (6.22) gives

$$\delta i(s) = \frac{s}{Ls^2 + (R_s + K_p)s + K_i}\, \Delta u_{\mathrm{ff}}(s). \tag{6.46}$$

The current response depends on the disturbance spectrum, the plant, and the controller gains. A static flux-error percentage cannot be converted into a current-error percentage without these additional quantities. At frequencies below the effective integral-control bandwidth, rejection is

strong; closer to the bandwidth, phase lag and coupling matter; beyond it, the feedback cannot follow rapid changes effectively.

This also separates two roles of reduced maps. Errors in $\boldsymbol{\psi}$ change predicted rotational voltage. Errors in its derivatives can change the true incremental plant relative to the design model and can also change scheduled gains if those derivatives are used for tuning. Feedforward error alone is an additive disturbance in the simplest analysis; model-dependent gain scheduling introduces an additional feedback-dynamics issue. Neither mechanism, by itself, proves that an observed oscillation was caused by a particular coordinate factor.

6.6 Voltage limitation and anti-windup

An inverter has finite voltage capability determined by its DC supply, modulation, and operating conditions. A useful circular approximation is

$$u_d^2 + u_q^2 \leq U_{\max}^2. \tag{6.47}$$

The precise conversion from DC voltage to $U_{\max}$ depends on modulation and scaling conventions. The circle is therefore an explicit actuator model, not a universal identity for every modulation regime.

At a specified feedforward voltage, the feasible PI correction set is

$$\{\boldsymbol{v} : \|\boldsymbol{u}_{\text{ff}} + \boldsymbol{v}\| \leq U_{\max}\}. \tag{6.48}$$

It is a disk centered at $-\boldsymbol{u}_{\text{ff}}$ in PI-voltage coordinates. Its shape makes direction important: the scalar margin $U_{\max} - \|\boldsymbol{u}_{\text{ff}}\|$ does not describe every possible correction equally. Moreover, if feedback compensates a map error exactly, the final total voltage is the physical voltage required by the machine. An erroneous feedforward does not necessarily increase that final physical requirement by its error norm. It can nevertheless alter transient commands and the allocation of limited voltage between feedforward and feedback.

The richer inspected limiter confirms a specific allocation order. It first reserves q-axis open-loop voltage, then d-axis open-loop voltage, then d-axis closed-loop voltage, and finally q-axis closed-loop voltage. A separate angular-control contribution receives the remaining capability with its own vector treatment. There is consequently a preference for d-axis feedback over q-axis feedback after the feedforward allocation. Describing the whole algorithm as a simple unconditional preference for d-axis total voltage would omit its first priority.

For example, after limiting the q-axis feedforward to the available magnitude, the algorithm computes the residual d-axis magnitude as

$$U_{d,\text{rem}} = \sqrt{U_{\max}^2 - u_{q,\text{ff,lim}}^2}. \tag{6.49}$$

It allocates d-axis feedforward within that residual, subtracts its absolute magnitude to form a conservative d-axis PI allowance, and then determines the q-axis PI interval from the circle and the already allocated components. This is a component allocation procedure; it need not coincide with radial projection of the total requested voltage. The chosen priorities explain why a q-flux or d-flux error can have different practical consequences even when their Euclidean magnitudes are equal.

Saturation also changes integral control. Without anti-windup, a persistent error continues to increase x even when further voltage cannot be applied. The stored integral then delays recovery after the command returns to a feasible region. A conventional back-calculation description is

$$\dot{x} = K_i e + K_{\text{aw}}(u_{\text{lim}} - u_{\text{req}}). \tag{6.50}$$

The second term drives the integral toward a value consistent with the applied voltage.

The richer inspected PI path forms the difference between requested and limited sums of feedforward and PI voltage, delays that difference by one update, and subtracts it from the integral input with gain K_i/K_p. With enabled control, constant step size T_s, and the nominal accumulator scaling, its essential structure can be expressed as

$$r_j[n] = u_{j,\mathrm{req}}[n] - u_{j,\mathrm{lim}}[n], \tag{6.51}$$

$$x_j[n] = x_j[n-1] + T_s \left(K_{i,j} e_j[n] - \frac{K_{i,j}}{K_{p,j}} r_j[n-1] \right). \tag{6.52}$$

The residual here concerns the main feedforward-plus-PI allocation; additional angular-control contributions have separate paths. Enable logic, gain adjustments, and numerical fallback behavior supplement this core relation. It establishes delayed back-calculation in the available structure, but does not establish its exact active parameterization in the measurements. Anti-windup improves recovery from saturation; it does not create additional actuator voltage or guarantee accurate tracking of an infeasible reference.

6.7 Operating regions and practical consequences

At low electrical speed, rotational flux-error voltage is small. This does not make flux modeling irrelevant: differential-inductance error, excitation-current changes, resistance uncertainty, and reference steps can still dominate the voltage mismatch. At standstill, a controller that supplies only rotational feedforward relies on feedback for resistive and transient voltage.

At high speed, the same flux error produces a larger rotational voltage error. Stable integral action may still remove the eventual current error, but the required compensation must develop through the closed loop and within the voltage limit. A claim of poor steady accuracy therefore requires evidence of saturation, instability, bias, or another limiting mechanism; speed scaling alone establishes increased voltage sensitivity.

During current steps, the flux derivative and the difference between requested and measured currents become important. Ideal instantaneous current steps require unbounded inductive voltage, so every physical current change has a finite achievable slope. A reduced map can affect both the predicted transient voltage and local gain scheduling. Its suitability for transients cannot be judged from stationary reconstruction error alone.

Near the voltage boundary, feedback has less freedom to correct a poorly predicted voltage. The verified allocation order can favor one correction over another, and anti-windup controls the evolution of the integral state during that restriction. The relevant comparison is consequently the direction and time history of the total voltage demand, not only the magnitude of the flux residual.

Finally, lookup evaluation and voltage generation need not have identical update requirements. In the normal reference-based path, unchanged lookup coordinates permit reuse of already evaluated flux values, provided map parameters and modes are also unchanged. Measured-speed multiplication can still require a fast update. In an alternative mode where measured currents enter the lookup, unchanged references do not imply unchanged arguments. Scheduled inductances, differential feedforward, excitation changes, and input selection must likewise be considered. Moving map evaluation to a slower or auxiliary task introduces age and synchronization errors that require a separate timing analysis. The available mathematical and reconstruction results establish no measured improvement in execution time or closed-loop performance from such an architectural change.

Chapter 7

Alternative Identification and Representation Strategies

The preceding chapters establish several different limitations: a contour statistic can depend on sampling, a common scalar coordinate can be incompatible with both flux components, a finite table can introduce its own errors, and measurement uncertainty can distort all of these diagnostics. A proposed improvement should state which limitation it addresses. Improving an estimator does not automatically increase the structural freedom of its coordinate model.

7.1 Robust contour identification

7.1.1 Length weighting changes the averaging measure

If an interpolated contour consists of segments with endpoints $\boldsymbol{a}_m, \boldsymbol{b}_m$ in the stator-current plane, define segment lengths $\ell_m = \|\boldsymbol{b}_m - \boldsymbol{a}_m\|$. On a straight segment, the coordinate integral is

$$\int_0^{\ell_m} \left(\boldsymbol{a}_m + \frac{\ell}{\ell_m}(\boldsymbol{b}_m - \boldsymbol{a}_m)\right) \mathrm{d}\ell = \ell_m \frac{\boldsymbol{a}_m + \boldsymbol{b}_m}{2}. \tag{7.1}$$

The centroid of the piecewise-linear curve is therefore

$$\boldsymbol{c}_{\text{length}} = \frac{\sum_m \ell_m(\boldsymbol{a}_m + \boldsymbol{b}_m)/2}{\sum_m \ell_m}. \tag{7.2}$$

Subdividing one straight segment without changing its geometry preserves this integral. This avoids the simple duplicate-point sensitivity of an unweighted crossing average. Disconnected components still require an explicit decision about whether their lengths should be pooled or their branches matched separately.

Length weighting does not force the resulting centroid onto a curved contour. The variance mechanism in Equation (3.20) remains, now with a different weighting measure. Noise in contour location, ambiguous branches and gaps in measured support remain relevant. It is therefore a more geometrically defined statistic, not a proof of unbiased physical factor identification.

7.1.2 Robust locations and fixed-q comparisons

A componentwise median or a geometric median can reduce the effect of isolated extreme crossing coordinates. For example, the geometric median minimizes $\sum_m \|\boldsymbol{c} - \boldsymbol{c}_m\|$. It does not necessarily lie on the contour, and changing the number or extent of disconnected pieces can still move

it. Robust statistics address a specified contamination model; they do not remove curvature or coverage effects without further assumptions.

Comparing at the same physical $i_q = q_0$ more directly addresses the interpretation of Equation (3.6). For each excitation level, solve

$$\psi_d(d, q_0, e) = \lambda \tag{7.3}$$

for a consistent d-current branch. Differences of these d-values no longer contain a moving-q term. Existence, branch uniqueness, sufficiently large L_{dd}, overlapping support and reliable interpolation are necessary. A family of nominal groups with measured excitation scatter also needs a clearly defined excitation coordinate at the selected crossing. Replacing measured excitation by a group label silently changes the experiment.

A different geometric approach aligns entire contours. For instance, one can minimize a weighted distance from the shifted first contour to the second:

$$\mathcal{E}(\delta) = \int_{\mathcal{C}_g} \operatorname{dist}((d+\delta, q), \mathcal{C}_{g+1})^2 \, w(d, q) \, \mathrm{d}\ell. \tag{7.4}$$

Turning the displacement into a factor additionally requires a defined excitation separation. Density weights, unmatched contour ends, ambiguous nearest points and branch changes affect the result. Symmetric two-direction distances can reduce one-sided matching bias, but do not supply missing contour coverage.

These alternatives are mathematical proposals here. The available conventional and diagnostic results motivate them; a new quantitative ranking of these contour estimators has not been performed in this manuscript.

7.2 An optimized global scalar factor

7.2.1 Reconstruction fitting and fixed-map alignment are different

For a candidate constant k, transform every training point by $(s, q) = (i_d + k i_e, i_q)$ and construct a matching common table $\widehat{\boldsymbol{F}}_k$. On a population $\mathcal{P}$ fixed before searching, a joint criterion is

$$\mathcal{J}_{\mathrm{rec}}(k) = \frac{1}{|\mathcal{P}|} \sum_{j \in \mathcal{P}} \left\| \widehat{\boldsymbol{F}}_k(i_{d,j} + k i_{e,j}, i_{q,j}) - \boldsymbol{\psi}_j \right\|^2. \tag{7.5}$$

Both components enter in Vs with equal weights. Replacing the mean by a sum gives the same minimizer when $\mathcal{P}$ is fixed. There is no component normalization that implicitly makes one axis more important.

The subscript k on the table is essential: the model is rebuilt consistently with the coordinates. In a fixed-map diagnostic, the reference $\widehat{\boldsymbol{F}}_{\mathrm{ref}}$ stays unchanged while only the query moves. That asks how well a specified reference can be aligned, rather than how well a newly constructed reduction performs. The pointwise diagnostics in Chapter 5 use such fixed references.

A finite grid makes Equation (7.5) nonsmooth in practice. Changing k can switch a scattered triangulation, a rejected simplex or a supported raster cell. A bounded search result is consequently a computed numerical candidate, not a proven global optimum.

7.2.2 Keeping the evaluation population fixed

Computing a mean only over whichever points each candidate happens to support creates a serious loophole: a candidate could improve its score by losing the difficult points. One remedy

is a predetermined population and an infeasibility condition whenever a candidate loses any of those predictions. The recent consistent-global fits use a one-grid-cell interior support guard at the reference seed to choose such a population.

A different existing investigation replaced missing components by a 1 Vs residual while keeping its evaluation denominator fixed. This also prevents simply omitting points, but changes the criterion strongly. In the earlier constant fixed-map two-axis diagnostic, the missing-data contribution accounted for approximately 99.6–99.7% of the total objective. Its selected factors primarily reflected the coverage trade-off. That diagnostic is therefore not presented as the practical consistent-global solution.

A fixed-population rule still requires reporting its population and the model's full coverage separately. It does not prove that excluded boundary regions are unimportant or validate extrapolation there.

7.2.3 What the existing scalar comparisons establish

A jointly fitted scalar and a contour factor can differ because they optimize different finite-data quantities, not because one is automatically a corrected physical turns ratio. The archived joint-scalar fitting used inner omitted-group queries rather than the recent all-data reconstruction objective. It also differs from the derivative criterion (3.46).

On the archived common interior prediction points, conventional scalar, jointly fitted scalar, two-axis and full three-current results are compared later in Table 7.5. Joint scalar fitting improves the first two datasets there but does not improve the third. Eliminating a contour-centroid statistic is therefore useful as a change of objective, but is not a universal solution of the reduction problem.

7.3 Two constant global excitation factors

7.3.1 The complete local condition

Consider one common coordinate pair for both flux outputs:

$$s_d = i_d + k_d i_e, \qquad s_q = i_q + k_q i_e, \qquad \boldsymbol{\psi} = \boldsymbol{F}(s_d, s_q). \tag{7.6}$$

The two constants form $\boldsymbol{k} = (k_d, k_q)^{\mathsf{T}}$. At fixed (s_d, s_q), the physical stator currents satisfy

$$\frac{\mathrm{d}}{\mathrm{d}i_e}\begin{bmatrix} i_d \\ i_q \end{bmatrix} = -\begin{bmatrix} k_d \\ k_q \end{bmatrix}. \tag{7.7}$$

For each output, differentiate along this discarded direction:

$$\left.\frac{\partial \psi_d}{\partial i_e}\right|_{\boldsymbol{s}} = L_{de} - L_{dd}k_d - L_{dq}k_q, \tag{7.8}$$

$$\left.\frac{\partial \psi_q}{\partial i_e}\right|_{\boldsymbol{s}} = L_{qe} - L_{qd}k_d - L_{qq}k_q. \tag{7.9}$$

The necessary condition for exact reduction is therefore

$$\boxed{\begin{bmatrix} L_{de} \\ L_{qe} \end{bmatrix} = \begin{bmatrix} L_{dd} & L_{dq} \\ L_{qd} & L_{qq} \end{bmatrix}\begin{bmatrix} k_d \\ k_q \end{bmatrix}.} \tag{7.10}$$

This uses all six derivatives of the two measured flux outputs with respect to the three currents. Excitation-circuit flux and its three additional derivatives have not been independently identified here.

The scalar model restricts the compensation to $(k, 0)^{\mathsf{T}}$. A nonzero k_q adds a second direction through which excitation can be compensated. This differs from fitting separate d-shifts for the two outputs: both outputs in Equation (7.6) use the same coordinate pair.

7.3.2 Local solutions, global constants and invertibility

If $\det \boldsymbol{L}_s = L_{dd}L_{qq} - L_{dq}L_{qd} \neq 0$, matrix inversion gives the unique local compensating pair

$$k_d^{\mathrm{loc}} = \frac{L_{qq}L_{de} - L_{dq}L_{qe}}{\det \boldsymbol{L}_s}, \tag{7.11}$$

$$k_q^{\mathrm{loc}} = \frac{L_{dd}L_{qe} - L_{qd}L_{de}}{\det \boldsymbol{L}_s}. \tag{7.12}$$

A small determinant relative to matrix size makes this computation sensitive; conditioning must be checked, not just algebraic nonzero values. For singular $\boldsymbol{L}_s$, $\boldsymbol{b}$ may be outside its range, giving no exact local pair, or inside its range, giving nonunique pairs.

Existence of a local pair at every point does not establish that one constant pair works everywhere. With a prescribed constant $\boldsymbol{k}$, the augmented transformation

$$\begin{bmatrix} s_d \\ s_q \\ i_e \end{bmatrix} = \begin{bmatrix} 1 & 0 & k_d \\ 0 & 1 & k_q \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_e \end{bmatrix} \tag{7.13}$$

is always invertible and has determinant one. Discarding its third coordinate remains a projection: distinct physical points can share the same (s_d, s_q). Equation (7.10), together with the corresponding connected-fibre or intercomponent consistency condition, is locally and globally interpreted exactly as in Proposition 3.1. Invertibility of $\boldsymbol{L}_s$ and invertibility of the augmented coordinate change are different questions.

Likewise, the pointwise inverse problem in Chapter 5 can fit one known flux pair with two query coordinates. That does not require every point to share a common physical compensation direction, nor does it identify an online function of measured currents alone.

7.3.3 Consistent global fitting and current results

The recent two-axis global fit rebuilds the common 81×81 table for each candidate. It minimizes the equal-component squared flux error on the fixed populations 7093, 7693 and 8625 for Sets 1–3. Any candidate losing a member of that population is infeasible. Bounds are centered on the available reference pair, with several targeted starts and a finite search budget. These settings define a focused local numerical refinement.

Table 7.1 gives the resulting all-data factors and own-supported-point errors. The reference pairs used for pointwise two-axis alignment were instead $(14.336652, -0.025602)$, $(14.186955, -0.091824)$ and $(14.627088, -0.787874)$ A/A from the earlier common-map analysis. Their tables remain fixed for that diagnostic; they are not silently replaced by the new maps. The matched four-method reconstruction comparison in Chapter 5 uses a different, explicitly shared population. Neither table is an independent prediction test.

Table 7.1: Consistent global two-axis fits. Errors are reconstruction RMSE in mVs on each fit's own supported points, not the smaller pointwise common populations.

Set	k_d	k_q	RMSE d	RMSE q	Joint	Valid/total
1	14.334062	−0.013622	5.037	4.198	4.636	7213/7476
2	14.302419	−0.078626	5.175	4.236	4.729	7809/8092
3	14.826705	−0.879645	5.370	5.502	5.437	8771/9022

7.3.4 Measured excitation scatter can fold a projected group

For exactly constant measured i_e, Equation (7.6) is a translation of each stator-current mesh. It preserves its signed triangle areas. A nominal group with an excitation graph $i_e = e(i_d, i_q)$ instead has Jacobian

$$J_{\text{graph}} = \begin{bmatrix} 1 + k_d e_d & k_d e_q \\ k_q e_d & 1 + k_q e_q \end{bmatrix}. \tag{7.14}$$

Expanding its determinant cancels the product term:

$$\det J_{\text{graph}} = (1 + k_d e_d)(1 + k_q e_q) - k_d k_q e_d e_q \tag{7.15}$$

$$= 1 + k_d e_d + k_q e_q. \tag{7.16}$$

A negative determinant reverses local orientation. A small determinant indicates local area compression. Slight pointwise excitation scatter can matter strongly on narrow or poorly conditioned current triangles because its estimated spatial gradient can be large.

For a triangle on which the measured excitation is linearly interpolated, this relation describes the exact affine projected area ratio. The numerical geometry check nevertheless compares the actual original and transformed triangles and retains their original connectivity. No artificial links between groups are needed to produce a fold within one group. Overlap between different excitation groups is also not, by itself, an error: overlap is required if a common map is to serve them. Their demanded flux values must agree there for exact reduction.

7.4 A general linear four-parameter transformation

7.4.1 Deriving the effective factors

The investigated extension is

$$\boldsymbol{s} = A\boldsymbol{i}_s + \boldsymbol{b} i_e, \qquad A = \begin{bmatrix} 1 & k_{dq} \\ k_{qd} & 1 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} k_{de} \\ k_{qe} \end{bmatrix}, \tag{7.17}$$

where $\boldsymbol{i}_s = (i_d, i_q)^{\mathsf{T}}$. The first index identifies the target coordinate, the second the source current. Thus $s_d = i_d + k_{dq} i_q + k_{de} i_e$ and $s_q = k_{qd} i_d + i_q + k_{qe} i_e$.

Let $D = 1 - k_{dq} k_{qd}$. If $D \neq 0$, direct multiplication gives

$$A^{-1} = \frac{1}{D} \begin{bmatrix} 1 & -k_{dq} \\ -k_{qd} & 1 \end{bmatrix}. \tag{7.18}$$

Multiplying Equation (7.17) by this inverse yields

$$A^{-1}\boldsymbol{s} = \boldsymbol{i}_s + (A^{-1}\boldsymbol{b}) i_e = \boldsymbol{i}_s + \boldsymbol{k}^{\text{eff}} i_e, \tag{7.19}$$

with

$$k_d^{\mathrm{eff}} = \frac{k_{de} - k_{dq}k_{qe}}{1 - k_{dq}k_{qd}}, \tag{7.20}$$

$$k_q^{\mathrm{eff}} = \frac{k_{qe} - k_{qd}k_{de}}{1 - k_{dq}k_{qd}}. \tag{7.21}$$

Conversely, $\boldsymbol{b} = A\boldsymbol{k}^{\mathrm{eff}}$ gives $k_{de} = k_d^{\mathrm{eff}} + k_{dq}k_q^{\mathrm{eff}}$ and $k_{qe} = k_q^{\mathrm{eff}} + k_{qd}k_d^{\mathrm{eff}}$. This inverse relation provides a numerically transparent parameterization by two shears and two effective factors.

If a continuously selectable flux function is $\boldsymbol{G}(\boldsymbol{i}_s + \boldsymbol{k}^{\mathrm{eff}} i_e)$, define $\boldsymbol{F}(\boldsymbol{s}) = \boldsymbol{G}(A^{-1}\boldsymbol{s})$. Then

$$\boldsymbol{F}(A\boldsymbol{i}_s + \boldsymbol{b}i_e) = \boldsymbol{G}(\boldsymbol{i}_s + \boldsymbol{k}^{\mathrm{eff}} i_e). \tag{7.22}$$

An invertible A therefore adds a change of reduced coordinates, not additional structural freedom to remove excitation. Its four coefficients are not four independently identifiable magnetic compensation mechanisms.

7.4.2 Finite grids do not commute with the coordinate change

A regular rectangular grid in $\boldsymbol{s}$ maps to a generally oblique grid in $A^{-1}\boldsymbol{s}$. A finite interpolation operator on one rectangular grid does not generally give the transformed version of the interpolation on the other grid. Range normalization can also change the scattered triangulation and its long-simplex rejection. Raster alignment, finite support and interpolation can thus change reconstruction error even at identical effective factors.

The investigation uses the same 81-node resolution per reduced axis and the same linear scattered-field and multilinear-return conventions as the two-axis fit. Shears are bounded by ± 0.35, $\det A$ must remain at least 0.5, and its condition number must not exceed 3. The effective-factor bounds and fixed physical fit population come from the two-axis reference. Geometry constraints prevent new or deepened folds, and finalists with additional positive-area self-overlap are rejected. Existing measurement-graph folds are retained rather than smoothed away.

Table 7.2 reports the best computed candidates. Their effective factors agree, to the displayed precision, with the two-axis factors. Several starts, including starts with slightly changed effective compensation, were explored. Some four-dimensional starts exhausted their evaluation budget, and different starts gave different nearby solutions. No global optimality or unique physical parameter interpretation is established.

Table 7.2: Four global coefficients and their two effective factors, in A/A. The order is target then source.

Set	k_{dq}	k_{de}	k_{qd}	k_{qe}	k_d^{eff}	k_q^{eff}
1	−0.015959	14.334280	−0.000758	−0.024482	14.334062	−0.013622
2	0.006774	14.301887	0.000979	−0.064618	14.302419	−0.078626
3	0.029437	14.800811	−0.037396	−1.434100	14.826705	−0.879645

7.4.3 Measured benefit, coverage and a simple prediction check

Table 7.3 compares reconstruction on identical physical points. These are in-sample results after the finite table has been built. Own support changes from 7213 to 7212 of 7476 points, from 7809 to 7808 of 8092, and from 8771 to 8783 of 9022. None of the 7093, 7693 and 8625 fixed fit points is lost. Coverage is therefore approximately unchanged; the small reconstruction benefit is not obtained by dropping members of the fit objective.

Table 7.3: Matched-support joint reconstruction RMSE in mVs for two-axis and four-parameter models. These populations differ from the pointwise comparisons.

Set	Common points	Two-axis	Four-parameter	Reduction
1	7205	4.638689	4.610750	0.60%
2	7805	4.730083	4.707679	0.47%
3	8749	5.441288	5.414354	0.49%

A controlled shear-only fit holds the original two effective factors fixed. These controls use their own common fixed-fit populations of 7093, 7693 and 8625 points, rather than the larger populations in Table 7.3. On those control populations, the shear-only fits reproduce the complete selected improvement; removing the shear while retaining the same effective factors returns the original two-axis result. This directly supports a finite-representation interpretation for the observed benefit. It does not prove that no other bounded candidate could perform better.

The coordinate matrices remain well conditioned, with condition numbers approximately 1.017, 1.008 and 1.008. The original refined two-axis solutions have 129, 128 and 80 reversed projected triangles, respectively; the selected four-parameter solutions retain these counts, with no degenerate triangles and no additional positive-area self-overlap. An invertible coordinate matrix does not make the original three-current projection injective or erase the measured excitation graphs' folds.

A further existing check omits one inner nominal excitation group per dataset and refines the models using the remaining groups. Table 7.4 uses common supported points in those omitted groups. The benefit is small in Set 1 and reverses in Sets 2 and 3. The check does not constitute

Table 7.4: A single omitted-group check per dataset. Joint flux RMSE is in mVs; excitation references are in A. This is not the archived multi-group prediction protocol.

Set	Omitted group	Common points	Two-axis	Four-parameter
1	20	566	3.942223	3.923104
2	17	566	4.089728	4.205774
3	15	911	4.547489	4.830587

a blind independent measurement campaign or remove earlier design choices informed by the datasets. It supplies no consistent prediction advantage. The additional focused fitting took roughly 30–50 s per dataset in the inspected runs, compared with roughly 11–13 s for the two-axis refinement. These are observed offline times, not target-controller timing measurements.

7.5 Full three-dimensional flux maps

7.5.1 What retaining excitation removes, and what it does not

A full representation uses

$$\widehat{\boldsymbol{\psi}} = \widehat{\boldsymbol{F}}(i_d, i_q, i_e). \tag{7.23}$$

It does not require excitation sensitivity to lie in one constant compensable direction, or even in a common constant two-axis direction. It still approximates an empirical current-only relation; temperature variation, magnetic history and measurement distortion do not disappear by adding the excitation axis.

A regular-grid lookup uses eight surrounding corners per component for trilinear interpolation, rather than four for a bilinear two-coordinate table. This corner count is not a hardware

execution-time measurement. The measured support must still be respected; a rectangular array can contain unavailable nodes inside its bounding box.

The stored full tables in the existing comparison contain $81 \times 81 \times 21 = 137781$ nodes, versus $81^2 = 6561$ for each reduced map. For two double-precision flux arrays, the value storage is

$$B_{2D} = 2(81^2)(8) = 104976 \text{ bytes}, \tag{7.24}$$

$$B_{3D} = 2(81^2)(21)(8) = 2204496 \text{ bytes}. \tag{7.25}$$

These are approximately 0.105 and 2.20 decimal MB, before masks and axis vectors. Twenty-one excitation grid nodes do not imply twenty-one independent measured excitation groups. They are storage locations sampled from the scattered field.

7.5.2 What the archived prediction results show

An existing whole-group prediction investigation removed each nominal excitation group in turn, refitted its model from the remaining groups, and assessed the omitted group. For fitted coordinate models, training contained smaller inner-group queries with a fixed missing-prediction penalty; the full three-current representation used the corresponding training measurements. The numerical choices and candidate hierarchy were fixed within that investigation. Its results are reused here without rerunning that campaign.

Table 7.5 selects the conventional scalar, jointly fitted scalar, two-axis and full three-current results on the same interior prediction points. The two-axis coefficients and fitting objective in this archived comparison differ from the recent fixed-population reconstruction refinement. Interior means that a nominal excitation group has retained groups on both sides. Pooled

Table 7.5: Archived common-support interior whole-group prediction. Joint RMSE in mVs. Each column uses one common physical prediction population across the complete archived comparison, not a mask selected for one displayed pair.

Representation	Set 1	Set 2	Set 3
Conventional scalar	5.998	6.309	6.970
Jointly fitted scalar	5.518	5.736	6.996
Two global axis factors	5.490	5.653	6.529
Full three-current map	1.787	1.780	1.480
Common prediction points	5659	6175	6592
Eligible interior points	6348	6936	7264

errors sum squared point errors before taking a square root; they are not arithmetic means of per-group RMSE. Boundary excitation groups have a different support problem and are not silently included.

The full representation is substantially more accurate on these supported points, but this is not a storage-matched comparison and does not identify a physical irreducible error floor. The full table is larger, its interpolation geometry differs, and all models use uncertain source measurements. When a boundary group is removed, a full three-current query can lose all support because the remaining excitation range no longer surrounds it. Reduced coordinates can remain numerically supported even then, but that is not a general guarantee of excitation extrapolation.

Existing grid-resolution and coordinate-scaling controls confirm numerical sensitivity. Consequently, neither the gain from a full table nor the difference between two RMSE values can be assigned exclusively to structural reduction error. Retaining excitation is a meaningful reference and a practical option when memory permits; it is not exact ground truth.

7.6 Auxiliary computation and reduced online workload

7.6.1 Identify the actual dependencies before caching

A larger map might be evaluated outside the fastest current-control update while retaining fast PI feedback and voltage limiting. This is an architectural possibility, not a measured real-time improvement in the present evidence.

The relevant input vector must be traced through the chosen operating mode. In the richer inspected controller, normal lookup coordinates use requested stator and excitation currents, whereas a mode selection can substitute internal measured stator currents. The simpler available structure also constructs its coordinates from references. These examples do not prove that every lookup in every mode depends exclusively on unchanged references.

Even if flux lookup inputs remain constant, rotational feedforward $\omega_e J\widehat{\boldsymbol{\psi}}$ changes when speed changes. Resistance feedforward can change with temperature or its resistance estimate. Gain calculations can depend on resistance, dynamics factors and multiplicative settings. One must distinguish caching a map value from caching an entire voltage command or controller state.

7.6.2 A conditional bound for stale map values

Let $\boldsymbol{\xi}(t)$ denote all inputs of a flux lookup, and suppose the last evaluation occurred at t_k. If the map is locally Lipschitz with bound M over the intervening supported region,

$$\left\|\widehat{\boldsymbol{\psi}}(\boldsymbol{\xi}(t)) - \widehat{\boldsymbol{\psi}}(\boldsymbol{\xi}(t_k))\right\| \leq M \left\|\boldsymbol{\xi}(t) - \boldsymbol{\xi}(t_k)\right\|. \tag{7.26}$$

If additionally $\left\|\dot{\boldsymbol{\xi}}\right\| \leq v_{\max}$, integration gives

$$\|\Delta\boldsymbol{\psi}_{\text{stale}}(t)\| \leq M v_{\max}(t - t_k). \tag{7.27}$$

At identical actual speed, the associated rotational voltage error is bounded by

$$\|\Delta\boldsymbol{u}_{\text{stale}}(t)\| \leq |\omega_e(t)| M v_{\max}(t - t_k). \tag{7.28}$$

The assumptions matter: no jump in reference, valid support throughout, and a usable slope bound are required. A lookup crossing a missing cell or a reference step is not covered by this smooth-input bound. High speed tightens any allowed flux-value age for a given voltage-error requirement.

For reduced coordinates, the input change already includes excitation:

$$\dot{s}_d = \dot{i}_d + k_d \dot{i}_e, \qquad \dot{s}_q = \dot{i}_q + k_q \dot{i}_e. \tag{7.29}$$

Detecting only stator-reference changes would miss a changing excitation reference. A nonlinear coordinate map would require its complete chain rule.

7.6.3 Scheduling, consistency and limits

An auxiliary evaluator would need a defined update period or event threshold, a maximum result age, a consistent input snapshot and an atomic transfer of both flux components. Derivatives or inductances used together should correspond to the same inputs and table version. Otherwise an apparently faster architecture can introduce incoherent feedforward and tuning data.

The fast loop would still use current measurements, update PI states, calculate instantaneous speed-dependent terms and enforce voltage constraints. A stale lookup is a delayed feedforward

contribution; it does not justify slowing feedback. Finite-difference derivative feedforward additionally needs a time base consistent with the held values. Simply reusing an old flux while differentiating a new reference can create a mismatch.

Worst-case execution time, scheduling jitter, memory access, data-transfer overhead and behavior during support loss would have to be measured on the actual target. No such real-time benchmark is available. A larger full map with a slower auxiliary lookup is therefore a candidate architecture whose accuracy-delay-cost trade-off remains open, rather than an established practical improvement.

7.7 Further possibilities and the boundaries of this evidence

A current-dependent coefficient introduces extra derivatives. If $s = i_d + i_e k(\boldsymbol{i})$,

$$\dot{s} = (1 + i_e k_{i_d})\dot{i}_d + i_e k_{i_q}\dot{i}_q + (k + i_e k_{i_e})\dot{i}_e. \tag{7.30}$$

Using the coefficient itself as a differential compensation slope would omit these terms. If a differential factor depends only on excitation, an accumulated shift instead satisfies

$$a(i_e) = \int_0^{i_e} k_{\mathrm{diff}}(\xi)\,\mathrm{d}\xi. \tag{7.31}$$

In general $a(i_e) \neq i_e k_{\mathrm{diff}}(i_e)$. More flexibility also raises inverse, regularization, support and identification questions; it does not turn target-informed pointwise factors into independently validated online functions.

Nonlinear coordinates, coefficient models, radial-basis representations and splines can change approximation and storage properties. Choosing a different basis is separate from choosing the input dimension. A flexible three-current fit does not prove that a two-coordinate reduction exists, and a flexible two-coordinate fit cannot represent two different flux pairs at exactly the same input. Preparatory work on such representations does not provide the missing comprehensive comparison here.

An energy-consistent representation could impose additional integrability and physical structure under appropriate assumptions. The present evidence does not independently identify the excitation flux and all associated derivatives needed for a complete three-current magnetic-energy model. These possibilities remain an outlook, not additional demonstrated solutions.

Chapter 8

Discussion and Conclusions

8.1 What the conventional procedure assumes

The conventional coefficient is the arithmetic mean of absolute-numerator d-current displacement quotients between adjacent nominal excitation groups at selected d-flux levels. Its coordinates are arithmetic averages of measured edge crossings. Its result therefore depends on the flux interval, group order, crossing distribution, edge filter and quotient weights. Nominal excitation determines grouping; measured excitation determines the coordinate differences and transformations.

Interpreting that statistic as a local magnetic compensation slope requires more than a clear contour plot. The local d-flux slope at fixed q-current is L_{de}/L_{dd} under its nonzero-denominator assumption. An arbitrary contour trajectory contains the additional q-motion term. A numerical centroid can also lie off the chosen contour, so even that trajectory equation need not describe centroid movement.

8.2 What is mathematically established

The curved-contour example establishes a centroid departure proportional to q-current variance even when mean q-current is zero. It also establishes that equal centroid departures in adjacent groups can cancel from the factor estimate. Differences of second moments, not centroid departure alone, determine the resulting signed displacement bias in that example.

Unweighted crossing means are not invariant to changes in crossing multiplicity or asymptotic sampling density on a geometrically unchanged curve. This is a mathematical weakness of their interpretation as a unique geometric contour location. It does not prove the magnitude of that weakness in each measured dataset.

For one constant scalar coordinate, exact common stator-flux reduction requires both excitation sensitivities to equal the same scalar times their d-current sensitivities. These differential identities are locally sufficient under differentiability. A global statement additionally requires consistency along complete fibres, including equality between disconnected fibre components when they occur. Pointwise collinearity with a varying local factor is weaker than one constant global reduction.

Two constant axis factors relax the restriction to d-compensation alone. Their local condition is $\boldsymbol{b} = \boldsymbol{L}_s\boldsymbol{k}$, but an invertible $\boldsymbol{L}_s$ gives only a local compensating pair, not a proof that the pair is constant everywhere. The augmented linear current transformation remains invertible while its two-coordinate projection can identify physically different points. These notions of invertibility must not be confused.

Finally, a regular four-coefficient coordinate matrix is equivalent to two effective excitation

factors followed by an invertible two-dimensional coordinate change. It cannot create additional continuous structural reduction freedom. A finite lookup table can nevertheless be sensitive to that change.

8.3 What the measured evidence establishes

The conventional coefficients for the three datasets are approximately 13.4921, 13.5184 and 14.7642 A/A. Their own-supported-point joint reconstruction RMSE values are approximately 5.03, 5.43 and 5.88 mVs. These values describe the complete finite-table return operation at measured points, not prediction at a new excitation level.

Nonzero centroid departures and q-current drift confirm that the conventional contour statistic is not simply a collection of fixed-q constant-flux samples. Nonzero estimated derivative residuals show imperfect local compensation in the empirical fields. Their magnitudes cannot be converted directly into flux-error percentages or assigned to a unique physical cause.

Pointwise scalar alignment improves an in-sample fit to a fixed reference but generally cannot match an arbitrary pair of flux components with one query degree of freedom. Two pointwise query coordinates can often approach the target pair much more closely. This improvement uses known target flux, and filters for weak excitation, nonuniqueness and conditioning explain the smaller retained pointwise population. It does not demonstrate an independently usable parameter model.

The recent consistent-global two-axis fits use a matching rebuilt table and a fixed objective population. They must be distinguished from the earlier fixed-map global diagnostic, whose large missing-data penalty dominated its score. The four-parameter fits preserve the fixed population and give approximately 0.5% improvement on matched reconstruction points. Controls on their own fixed-fit common populations reproduce the selected gains without changing the effective factors. A single omitted inner group becomes slightly better in one dataset and worse in two. This is insufficient evidence to justify routine additional fitting complexity.

The archived multi-group comparison favors retaining the excitation coordinate within its supported interior domain: joint prediction RMSE is approximately 1.48–1.79 mVs for the full representation, compared with roughly 5.49–6.53 mVs for two global axis factors on the same points. This advantage is empirical and includes a larger table and different interpolation geometry. It is not a proof of an immutable physical accuracy floor for every possible two-coordinate representation.

8.4 What follows for current control

A model-minus-true flux error produces rotational feedforward error $\omega_e J \Delta\boldsymbol{\psi}$ at identical arguments and speed. This term grows with electrical speed. Transient voltage prediction additionally depends on derivatives of the flux error; changing excitation can disturb the stator dynamics even at zero speed.

A stable unsaturated PI loop can compensate a constant additive voltage error with its integral state. Therefore a flux RMSE is not a direct current-error RMSE. Dynamic rejection, cross-coupling, measurement bias, update delays and voltage saturation determine the actual tracking consequence. Close to the voltage limit, extra feedforward demand can consume the reserve needed for feedback, and anti-windup controls the integral response to unavailable voltage.

The available controller examples verify that reference-based lookup, diagonal inductance tuning, optional differential feedforward, different voltage-limit structures and mode-dependent inputs

are real design possibilities. They do not establish the active configuration or delay of the measurement hardware. In particular, neither ideal magnitude-optimum tuning nor a simple overall d-axis clipping priority can be assumed for every inspected structure.

8.5 Practical choices and remaining questions

A contour statistic is useful when a simple reproducible baseline is required, provided its sampling-dependent interpretation is retained. Length weighting or fixed-q comparisons address specific weaknesses but still require a fair measured comparison. Joint global fitting aligns the criterion with two-output flux approximation; it does not remove the scalar model's structural constraint.

Two global axis factors offer modest additional complexity and a simple inverse coordinate relation. A four-coefficient shear is useful for studying finite-grid sensitivity, but the measured benefit here is small and not consistent in the simple prediction check. A full three-current representation is a strong supported-domain reference when storage permits. Auxiliary evaluation could reduce fast-loop work, but its benefit requires target timing and delay measurements rather than an offline assumption that lookup inputs never change.

The unresolved questions are concrete. Which measurement and magnetic-state variations explain the empirical derivative mismatch? How much do contour weighting and matched-q sampling change factor identification on the same supported regions? Which controller modes and gain settings were active, and how do map errors affect measured transients near voltage limitation? Can a larger representation meet worst-case real-time deadlines with a controlled update age? The existing evidence does not answer these questions completely.

The distinction to retain is between an exact factorization theorem, an approximate finite-data representation and a practically useful controller model. The first needs mathematical assumptions, the second needs explicit numerical support and validation populations, and the third additionally needs a verified operating context and real-time performance evidence. No single numerical factor or reconstruction plot supplies all three.